

PROJECT DESCRIPTION DOCUMENT

CLIMATE, COMMUNITY & BIODIVERSITY STANDARDS

The Nature Conservancy of Canada

DARKWOODS FOREST CARBON PROJECT

v. 1.6

<i>Darkwoods Forest Carbon Project</i>	July 11, 2013
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G1. GENERAL SECTION

G1a. Original Conditions in the Project Area

The Nature Conservancy of Canada (NCC) acquired the fee simple 54,792 ha (135,394 acre) Darkwoods property near Creston, BC from the Pluto Darkwoods Corporation in April of 2008, with the objective of managing the land for ecological conservation objectives. The Darkwoods Forest Carbon Project was developed and Validated and Verified to the Verified Carbon Standard (VCS), Methodology VM0012 as a 100 year Improved Forest Management (IFM) – Logged to Protected Forest (LtPF) carbon project on April 21, 2011.

The Darkwoods Forest Carbon Project is management of the property for long term conservation of ecological function and values through the protection of the forest and various conservation-based management activities. The Darkwoods project achieves net GHG emission reductions and removals through the avoidance of emissions due to conventional logging in the baseline scenario. The management of the property for conservation purposes does not generate material levels of net revenue, and the sale of carbon offsets are critical to funding significant ongoing property conservation management and ownership costs.

Darkwoods Carbon Project Overview:

As determined in the VCS VM0012 baseline selection process, the most plausible baseline scenario is based on the pre-existing 3rd party certified appraisal of the property that drove the valuation of the property for the sale. In British Columbia, as in many jurisdictions, an independent expert appraisal process is typically used to determine a property valuation that drives the asking price, negotiation, and transaction value. The appraisal anticipates typical return-generating activities from the property asset from discounted cashflow analysis and comparable property valuations. The appraisal is designed to represent an acquisition price based on typical practices and typical financial return opportunities from comparable properties and appraiser experience, and hence provides an excellent independently developed estimation of baseline scenario activities. For Darkwoods, this appraisal baseline scenario outlined a property valuation based on a 15-year depletion of currently mature timber stocks on the property to target a typical North American timberland internal rate of return on investment. Under the this selected baseline scenario, a 100 year harvest schedule was developed to model the 15 year appraisal depletion period followed by uneven harvesting of the property based on the age class distribution growing to harvest age over time. The baseline scenario is modeled to follow the typical regional practice of clearcut logging with minimum legal requirements for private forestlands in B.C. and as demonstrated in comparable regional practices and adjacent properties.

In contrast, the project scenario is conservation management activities, wherein NCC undertakes the carbon project and ecosystem protection and enhancement activities. The project scenario implements and monitors forest protection across the majority of the property along with an ongoing low level of timber harvesting as part of conservation management activities for ecosystem/habitat enhancement, natural disturbance risk management, and long term age class structure development. These project scenario harvesting and related property management operations also provide an ongoing support to the local forest economy.

In addition to the carbon emission reductions demonstrated as part of the VCS project Validation and Verification, the Darkwoods Forest Carbon Project involves substantial net benefits to biodiversity and community socio-economic well-being, and hence the project is expanding to pursue Climate, Community, and Biodiversity Association (CCBA) Validation.

A high level of biodiversity protection and enhancement is a primary function of the project, with this large and diverse property providing important habitat for multiple at-risk and keystone species along with the protection and re-development of large scale fully functioning natural ecosystems.

The project area has no communities within the project area, and the project has no material impact on community value or use within the project area. The regional communities are very well diversified and economically robust. Although the project has a potential impact through reduction in logging employment opportunities for certain regional community members, the overall net impact of conservation of the property is positive, with net socioeconomic gains in ecosystem services provisioning value and improvement of key community well-being values (i.e. community amenity values), along with economic opportunities due to project harvesting, property management operations, monitoring, and other conservation management related economic activities.

The project will utilize the VCS PDD and Monitoring Report documentation as the primary source of project design information and all carbon accounting data and calculations for this CCB Validation. This CCB PDD will reference or re-state identical or similar information as found in the Validated VCS Darkwoods PDD and Verified VCS Darkwoods Monitoring Reports currently posted on the VCS website ([VCS Project Database - Darkwoods Project](#)). The CCB PDD serves to expand data and details related to the Biodiversity and Community sections of the project with respect to CCB Standards, Version 2 that are not fully described in the VCS PDD.

Darkwoods Project Background

Prior to the NCC acquisition, the Darkwoods property had been owned and managed by the Pluto Darkwoods Corporation on behalf of a His Royal Highness Duke Carl Herzog von Wurttemberg (a German aristocrat) since 1967. This previous management was focused on forest management with a low harvest level (averaging approximately 57,000 m³/year) and strong ecological and conservation management principles. For example, despite no legal requirement on private land and beyond even crown land regulations, the Pluto Darkwoods Corporation retained approximately an additional 1/3 of the merchantable timber areas on the property in generally protected condition¹.

After 40 years of ownership, the Pluto Darkwoods Corporation decided to divest the Darkwoods asset, apparently to re-deploy the asset capital and due to the operations not producing reasonable financial returns at the lower harvest levels. The history of conservation-based management on the property created retained critical biodiversity and habitat areas of interest to NCC; however, this previous management also resulted in a large component of mature merchantable timber standing stock that provides an important immediate revenue opportunity for market driven acquirers interested in acquiring the property. In addition, the property sits adjacent and overlooking Kootenay Lake, with potentially attractive real estate development opportunities in combination with the timber harvesting opportunities on the property. The key baseline scenarios described in Section G2 are based on independent 3rd party appraisals undertaken by certified B.C. Appraisers, and other evidence of parties interested in acquiring the Pluto Darkwoods property for these logging and development opportunities.

The property was offered for a sealed bid sale, first to a selected group of bidders, of which NCC was one. NCC viewed the Darkwoods property as under immediate threat of heavy logging and other industrial logging practices and/or extensive real estate development. Recent intensive logging activities on near by properties and other regional evidence of private forestland mature primary forest liquidation reinforced this threat assessment.

¹ Known as EPA's (Environmental Protection Areas) in the previous GIS inventory, these were areas of personal interest to the Duke that were voluntarily put off limits to timber harvesting.

In B.C., private land is lightly regulated in comparison to more heavily regulated Crown land requirements in B.C. and across Canada. Private rural residential land has little government oversight other than local land use zoning and general water quality protections, etc. when applicable. Private timberland in B.C. can be voluntarily registered as Private Managed Forest Land under the Private Managed Forest Land Act (PMFLA) in B.C. (administered by the Private Managed Forest Land Council (PMFLC), a B.C. crown entity), which provides a lower land tax rates in order to encourage landowners to retain the forest as forested land. The PMFLA has a focus on maximizing long term forest production with a set of best practices for providing minimum protections for key management objectives of public interest related to soil conservation, water quality, fish habitat, critical wildlife habitat, and reforestation. Generally, the PMFLA, PMFLC and the related advocacy B.C. Private Forest Landowners Association (PFLA) are highly supportive of private landowner rights and provide a set of basic minimum requirements and recommended best practices that do not have significant impacts on available timber harvesting area, restrict timber harvesting volumes or sequence, or strongly impact or restrict harvest operations. Intensive industrial logging within the minimum PMFLA requirements is legal and not uncommon in B.C. on private timberland. The forest management baseline scenarios described and modeled in detail in the VCS PD and this document exceed the minimum requirements in the PMFLA, as does the project scenario forest management. Note that the previous Pluto Darkwoods management was operated within the PMFLA, but was operationally not representative of either the legal requirements or common practice on comparable private lands due to the extensive conservation of merchantable timber area and other protective operating practices implemented by the private ownership.

G1.1 Location and Physical Parameters

The Darkwoods property is a 54,792 ha (135,394 acre) contiguous parcel of fee simple private property in south eastern British Columbia just north east of the municipality of Creston. The Darkwoods property is bounded by Kootenay Lake on the east and various crown and private land on the other property boundaries. There is a significant in-holding in the center of the property that is owned and managed by the Wynndel Box and Lumber Company, with access rights through the Darkwoods property. The boundaries are surveyed (see Table 1, Figure 1, and Figure 2) with the following legal land description:

Table 1 - Darkwoods Property Legal Descriptions and Parcel Identifiers

PID #	Description
PID 007-608-349	Sublot 1, District Lot 2381, Kootenay District, Plan X74, except (1) Plans
PID 007-608-594	Sublot 7, District Lot 2381, Kootenay District, Plan X74
PID 007-608-446	District Lot 887, Kootenay District, except part included in Plan 1760
PID 026-235-927	District Lot 15184, Kootenay District

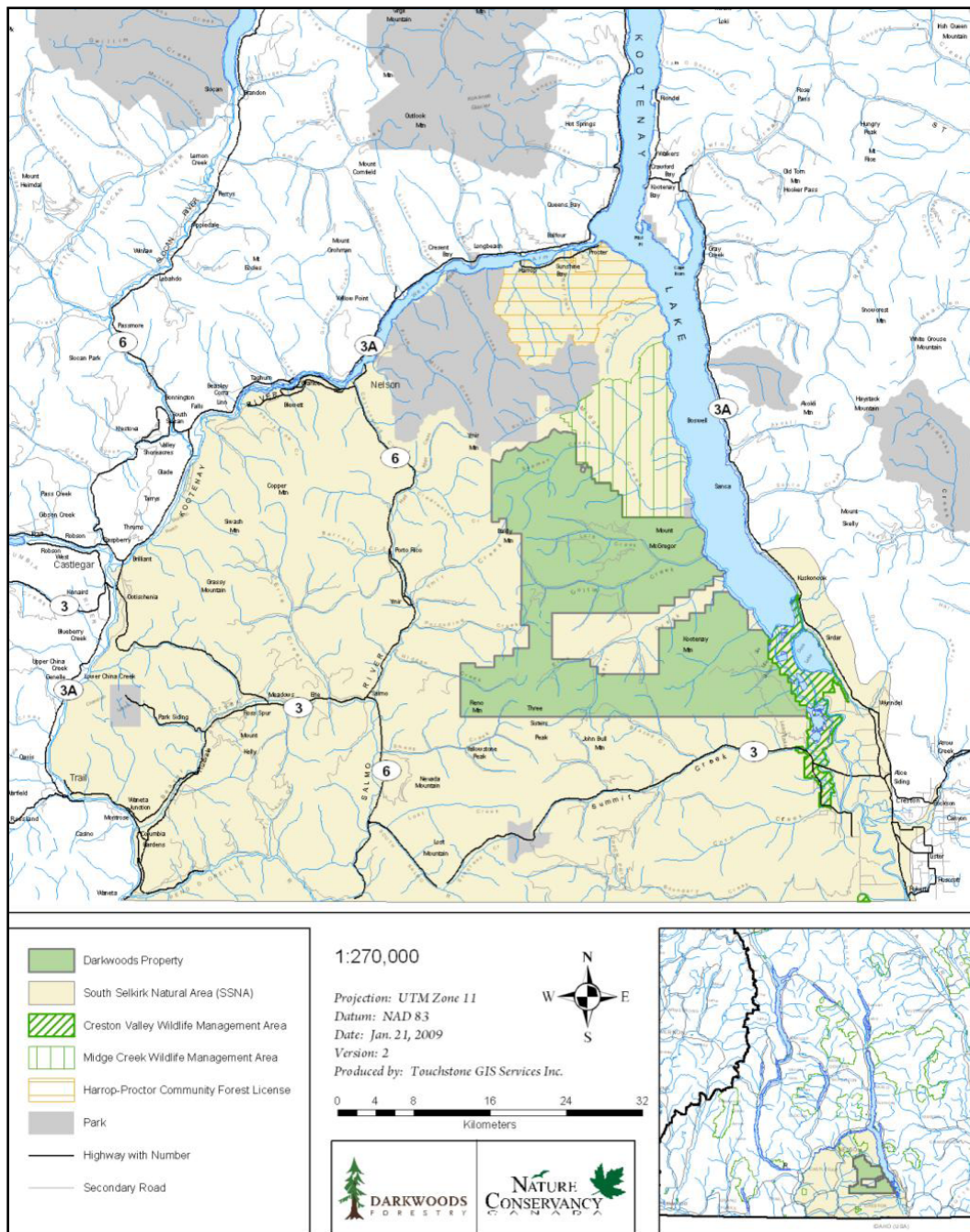
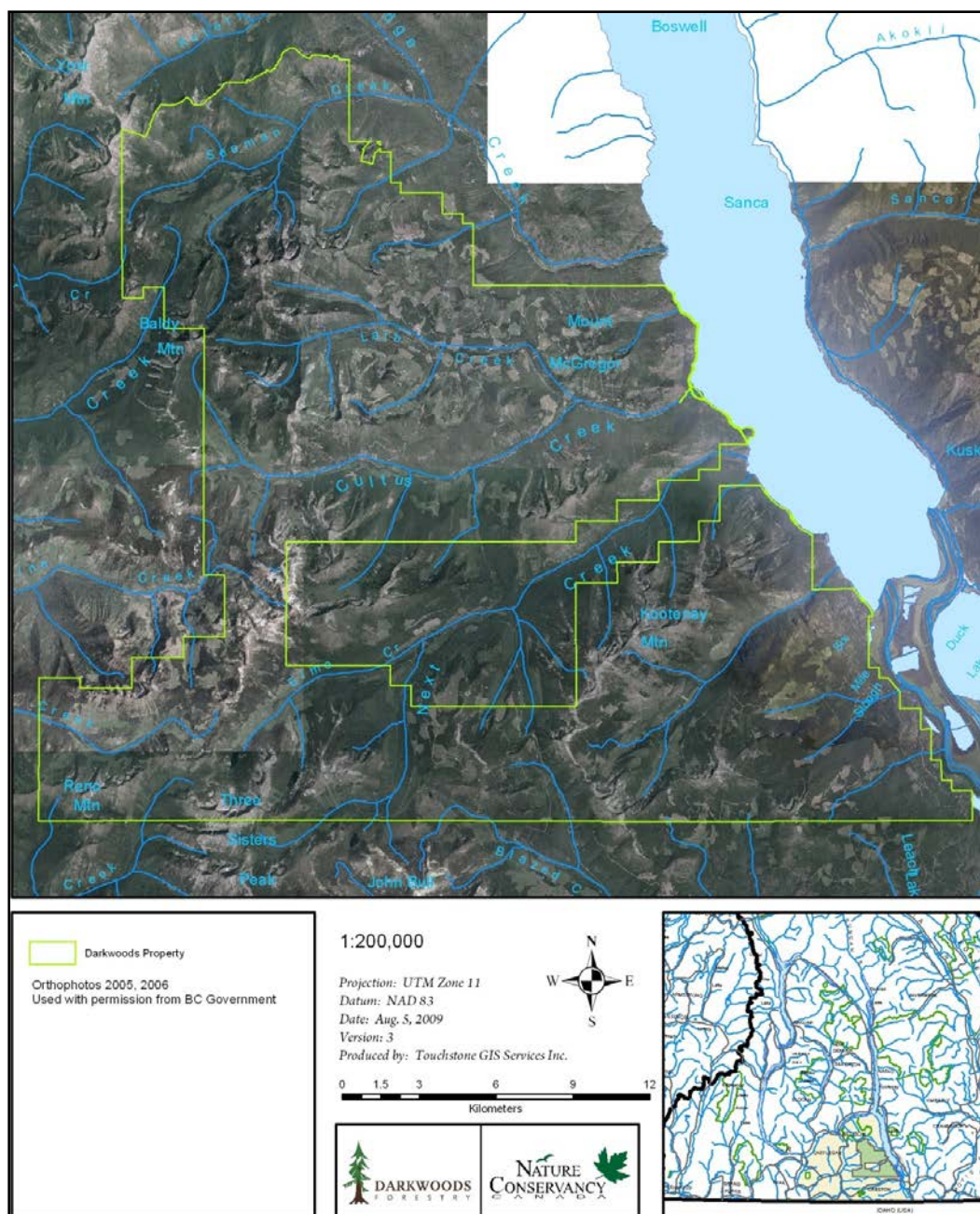


Figure 1 - Darkwoods Property Boundary and Project Area Overview Map



The following is a property overview summary from the draft Darkwoods Property Management Plan 2010-2015 (NCC, 2010):

The size of the property is 54,792 ha (135,394 acre). Forest harvesting, wild fire and forest pathogens have disturbed portions of the area over the past 100 years. Still, the current and long-term biodiversity potential is very high, and the property has extensive areas of natural origin primary forest. To date, 200 animal and 219 plant species have been confirmed on the property (Steeger and Machmer 2009).

Physical description: The property is located near the southern extent in BC of the Selkirk mountain range, which is locally bounded to the east by Kootenay Lake and to the west by the Salmo River valley. Elevations range from approximately 450 m in the Creston Valley to over 2400 m.

G1.2 Types and Conditions of Project Area Ecosystems

Biological description: Darkwoods is located in the Southern Interior Mountains ecoprovince, Northern Columbia Mountains ecoregion, and the Southern Columbia Mountains ecosection. Biogeoclimatic subzones include the Engelmann-Subalpine Spruce (ESSF) dry-mild (dm) and wet-cold (wc) subzones, as well as the Interior Cedar-Hemlock (ICH) dry-warm (dw) and moist-warm (mw) subzones.

IUCN red-list species: grizzly bear (*Ursus arctos*)(Threatened), wolverine (*Gulo gulo*)(Near Threatened), and bull trout (*Salvelinus confluentus*)(Vulnerable).

Other notable species: In total, 19 Globally, Nationally, and/or Provincially significant species-at-risk are predicted to occur on the property, 15 of which have been confirmed (Steege and Machmer 2009). Darkwoods likely provides significant habitat for at least 7 of these species-at-risk, including grizzly bear (Special Concern; blue list), mountain caribou (*Rangifer tarandus*) (Threatened; Red list), Olive-sided Flycatcher (*Contopus cooperi*) (Threatened; Blue list), Peregrine Falcon (*Falco peregrinus*) (Special Concern; Red list), western toad (*Bufo boreas*) (Special Concern; Yellow list), wolverine (Special Concern; Blue list) and bull trout (Blue list)(NCC 2011).

Legal status: Darkwoods is private land, owned in fee simple by the Nature Conservancy of Canada, subject to the encumbrances registered on title and the terms of the original Crown Grant. The management and disposition of the property is subject to NCC's Conservation Policy Framework as approved by its board of directors. Furthermore, the property purchase involved the Ecological Gifts Program at closing (and therefore implemented as part of, and in support of, the carbon project); therefore Environment Canada must approve any "Changes in Use" as defined by Environment Canada. The land is currently voluntarily registered as Managed Forest (Class 7) for assessment purposes, and as such NCC has obligations as a managed forest owner under the Private Managed Forest Land Act.

Historical description: The property was originally a crown land grant to the Nelson & Fort Sheppard Railway in 1897. It was held by various private interests until purchased by Nature Conservancy Canada in 2008 from the Pluto-Darkwoods Corporation, the latter of which had owned the property for the previous 40 years. The principal land use has been forestry.

Cultural description: Darkwoods is located within the traditional territory of the Ktunaxa First Nation. There has been a tradition of limited outdoor recreation on the property, including fishing, snowmobiling, skiing, and summer vacations at Tye (a recreational area with seasonal cabins on the property).

Access information: Permitted public access is provided through the Blazed Summit Forest Service Road or the Porcupine Creek Forest Service Road from the west near Salmo. Other access points include Wildhorse, Oscar, Hidden, Sheep, Nugget, Topaz and Newington roads. The residential Tye site can be reached by boat via Kootenay Lake. Other access to the property is often arranged by helicopter.

Regional Biogeoclimatic Overview and Context:

The following is an overview of the sub-regional bioclimatic conditions that relate to the project site (B.C. Ministry of Forests, 2008).

The Kootenay Lake Timber Supply Area (TSA) includes both moist and wet climatic regions, and is commonly referred to as part of the Interior Wet Belt. The moist climatic region covers most of the TSA, except for a wet region north of the Purcell Wilderness Conservancy (well to the north of the

Darkwoods property). Varied ecological features and species diversity contributes to the high biodiversity values in this TSA.

Three biogeoclimatic zones and four ecosections occur in the Kootenay Lake TSA. The Interior Cedar Hemlock (ICH) zone occupies valley bottoms and lower slopes to about 1400 meters. Four different subzones of the ICH occur in this TSA, reflecting differences in precipitation. They range from a drier subzone around the south end of Kootenay Lake where annual precipitation averages 70 cm to a wetter subzone in the Duncan Valley where annual precipitation averages 120 cm. In general, the ICH zone has wet, cool winters and warm, dry summers, and is the most productive forest zone in the interior of BC. The ICH has a high diversity of tree species including Western red cedar, Western hemlock, Grand fir, Engelmann spruce, Subalpine fir, Western larch, Douglas-fir, Western white pine, Western yew, Ponderosa pine and Lodgepole pine. The Engelmann Spruce-Subalpine Fir (ESSF) zone is the uppermost forested zone in the Kootenay Lake TSA, typically occurring at elevations between 1400 and 2500 meters (i.e., above the ICH and below the Interior Mountain-heather Alpine zone). The ESSF zone has a relatively cold, moist and snowy continental climate. Growing seasons are cool and short, while winters are long and cold. Engelmann spruce and Subalpine fir are the dominant climax tree species, while Alpine larch and Whitebark pine also occur. At the lower elevations of this zone, Lodgepole pine, Douglas-fir, Western hemlock and Western red cedar can be found. The Interior Mountain-heather Alpine (IMA) zone occurs at elevations greater than 2250 meters, above the ESSF zone. The climate is cold, windy and snowy with a short, cool growing season. By definition this area is largely treeless — consisting of rock, ice and snow. Shrubs, herbs, mosses and lichens are a common component in vegetated areas.

The diverse forests of the Kootenay Lake TSA support an abundance and wide variety of wildlife species. Large mammals include black bear, grizzly bear, moose, mule deer, white-tailed deer, cougars, elk, mountain goat, bighorn sheep and caribou. Mountain caribou require older forests for the arboreal lichens on which they depend for forage and for security cover, as well as large unfragmented forests for seasonal migrations. Seventy percent of the bird species known to occur in BC and 62% of bird species that breed in the province are known to exist in the Kootenay Lake area. More than 20 varieties of birds are area year-round residents of the TSA including golden eagles, grouse, woodpeckers, jays, magpies, ravens and English sparrows. The area also contains one of the highest breeding concentrations of ospreys in the world (B.C. Ministry of Forests, 2008).

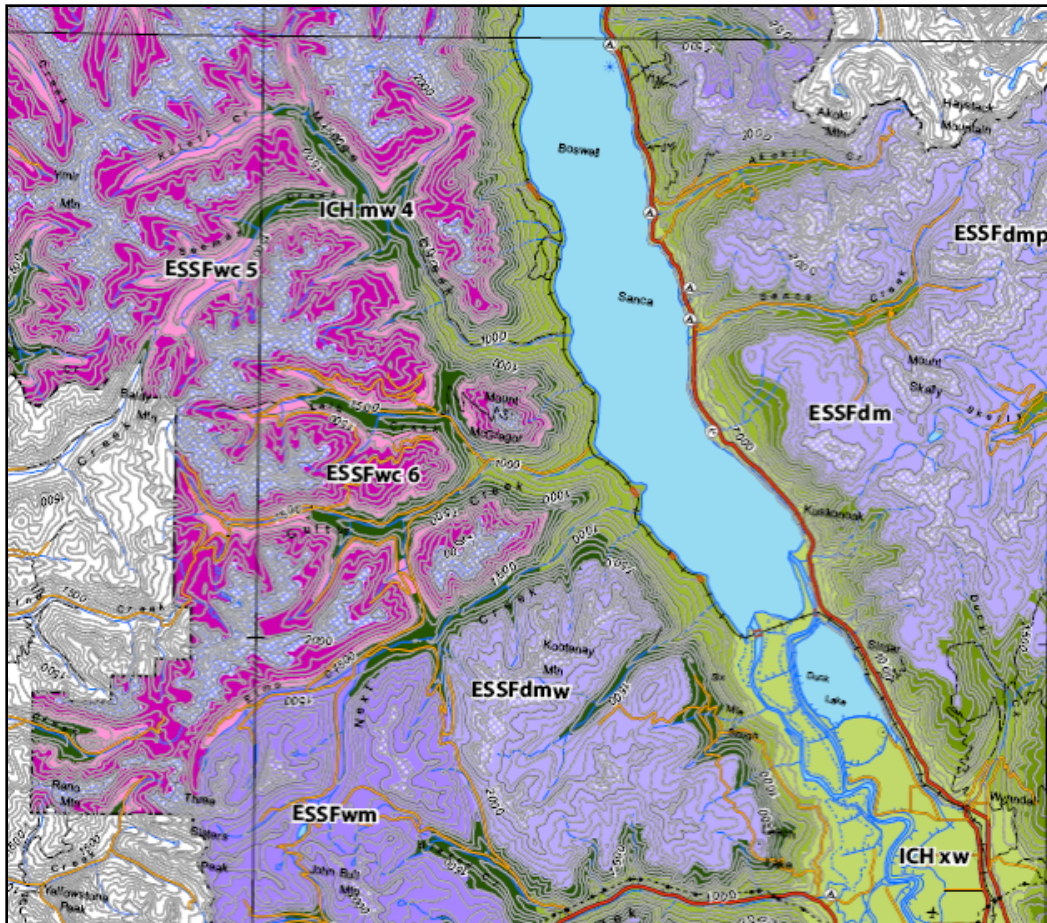


Figure 3 - Darkwoods Area Biogeoclimatic Subzone/Variant Field Map
<http://www.for.gov.bc.ca/HRE/becweb/>

Current State of the Property

The Darkwoods property has a diverse mix of forest types, topography, age class, and forest conditions well distributed across the landbase (Figure 3, Figure 4, and Figure 5). No single forest/site type accounts for more than 10% of the area (Table 2).

Approximately 68% of the property (37,250 ha) is considered operable for timber harvesting, as shown in Table 2. Within this operable area, approximately 9,012 ha are in managed/reforested stands <40 years old, which reasonably reflects the total area harvested and reforested during modern silvicultural practices. The balance of the operable area reflects natural and older harvesting (>40 years ago) which are reasonably assumed to have been regenerated naturally, and for all intensive purposes will behave similar to natural forest conditions.

The property is well-roaded, with main haul routes located into each drainage area, as shown on Figure 2 and Figure 4. NCC has completed a Property Management Plan that includes a road and access management program; which includes access management controls (i.e. gating), road stabilization, road deactivation, and permanent closure of certain roads and watersheds on the property. NCC project activities and harvesting has occurred annually since acquisition, as described in VCS Monitoring Report documents for 2008-2010, and 2011-12. The last project harvest activity was undertaken in 2012. Conservation planning and management continues on an ongoing basis.

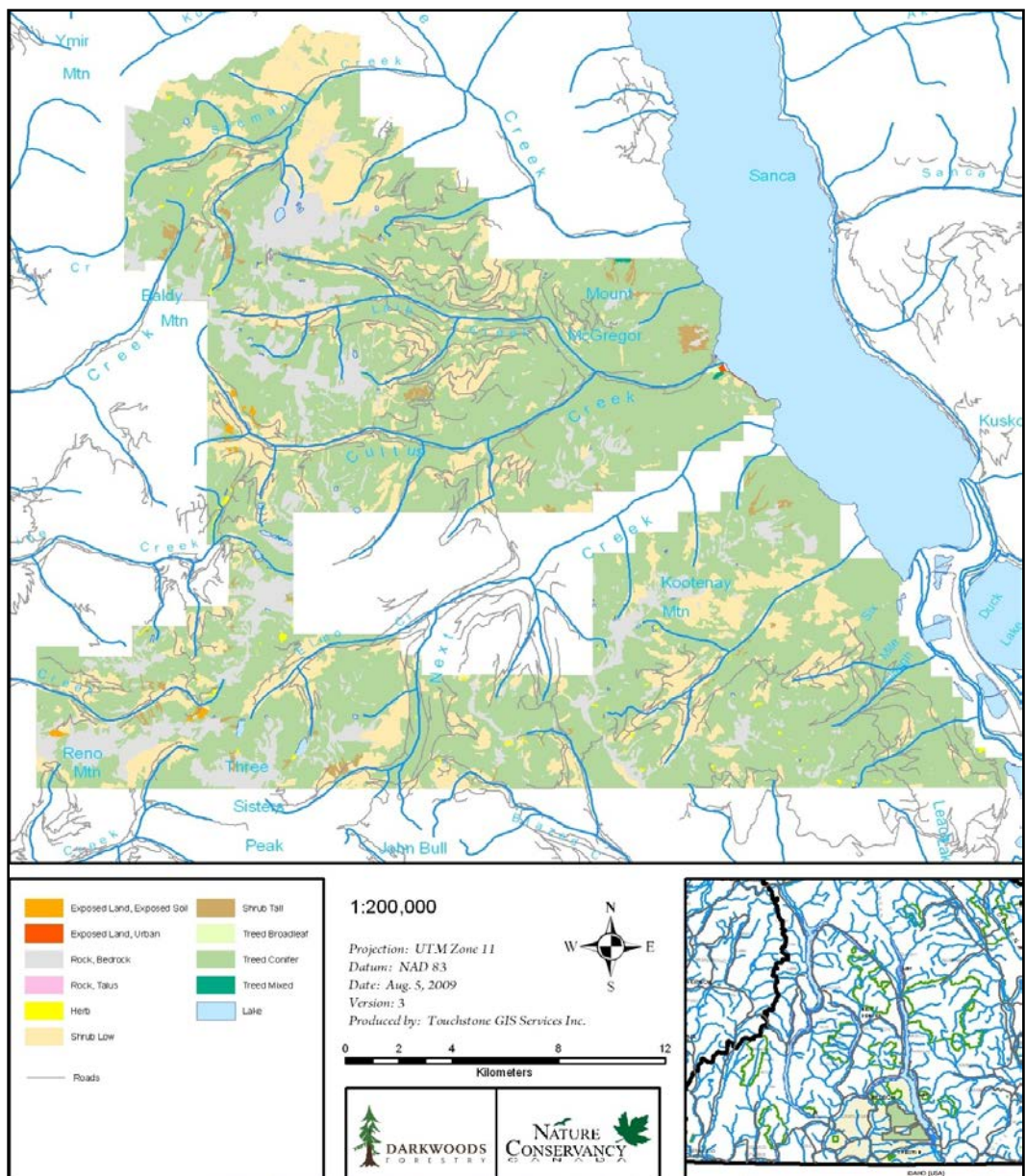


Figure 4 - Darkwoods Land Cover Map

Table 2 - Darkwoods Forest Cover Description by Leading Species, Productivity Class, Age Group, and Operability²

Description	Lead	Other	Productivi	SI*	Age	Operab	Inoperab
non-forest						0.0	7018.6
F / L_med	F / L	Pl	med	13-	>40	1909.7	419.4
F / L_good	F / L	H & C	good	>17	>40	2589.8	91.0
B / L / F_poor	B / L /	S, Pl	poor	<13	>40	7945.2	7742.8
B_med	B	S,Pl	med	>=1	>40	5202.2	294.5
P_poor	P	B, S	poor	<13	>40	1839.3	489.9
P_med	P	F, S	med	>=1	>40	2601.7	557.7
C / H_med-	C / H	S,F	med-good	all	>40	2333.4	78.6
S_med	S	Bl, Pl	med	<=1	>40	2778.0	240.2
S_good	S	F, H, C	good	>16	>40	1023.2	9.2
E / A_Med	E / A		Med	all	>40	15.3	0.0
F / L_med	F / L	Pl	med	13-	<=40	119.7	0.0
F / L_good	F / L	Pl	good	>17	<=40	326.5	1.2
S/P/B poor	S/P/B	L	poor	<13	<=40	2068.8	586.1
S/P/B Med	S/P/B	L	med	>=1	<=40	3615.0	4.1
C / H_med-	C/H	F, S	med-good	all	<=40	231.9	7.6
S_good	S	F, H, C	good	>16	<=40	2649.8	1.5
					Total	37,250	17,542

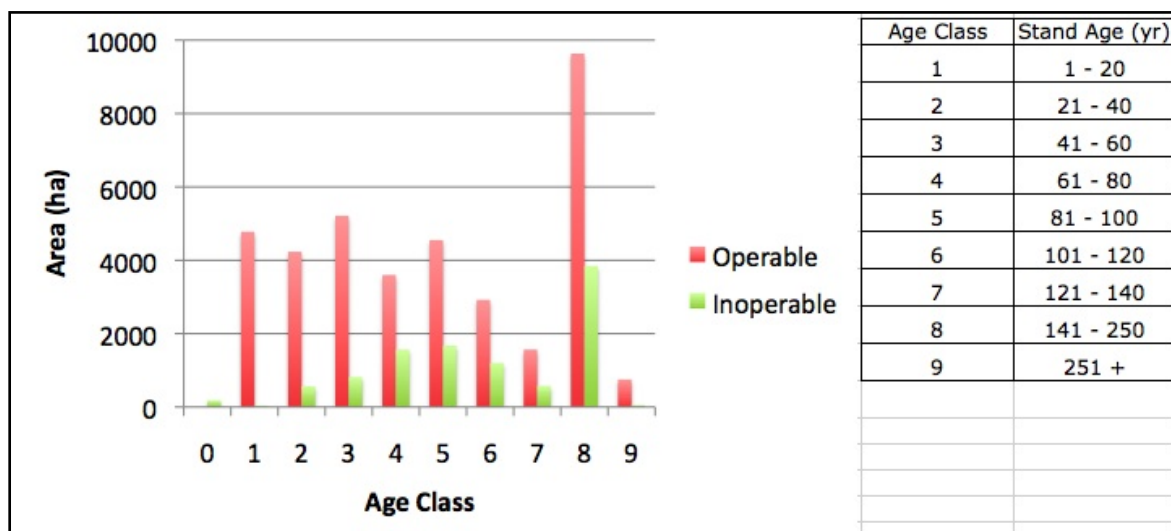


Figure 5 - Darkwoods Forest Area by Age and Operability (2008)

² F = Douglas-Fir, L = Western Larch, B = Subalpine Fir, S = Hybrid spruce or Engelmann Spruce, P = Lodgepole pine and White bark pine, C = Western red cedar, H = Western hemlock, E = Paper birch, and A = Trembling aspen.

G1.3 Project Area and Project Zone Boundaries

Project Area Boundaries: see Figure 1, Figure 2, Table 1.

Project Zone Boundaries: see Figure 6

G1.4 Climate Information

The Darkwoods project area sequesters significant carbon stocks in the forest biomass located across the property. The avoidance of baseline scenario logging results in the annual retention of carbon stocks by avoiding the removal of tree biomass and subsequent decay of the resulting dead biomass and wood product waste. The site also continues to sequester additional carbon in the growth of the retained and regenerating forests across the site. These carbon stocks and their fluxes overtime have undergone detailed measurement and modeling by carbon pool, using the Verified Carbon Standard (VCS) Methodology VM0012, as documented in the Validated Darkwoods Forest Carbon Project Design Document (PDD).

The starting and current status of carbon stocks are reported in:

- Darkwoods VCS Project Design Document (3GreenTree, 2011), Version 1.8. Validated April 21, 2011 (Rainforest Alliance, 2011a). The Darkwoods PDD documents all project design details, carbon stock calculations, risk assessments, and projects ex-ante VCU carbon offset for the duration of the project. Key information related to carbon stock calculations include:
 - Section 2.3 for GHG pools included
 - Section 4 for Baseline and Project Scenario carbon accounting calculations
 - Figure 11 summarizes the starting ecosystem carbon stock and initial PDD ex-ante projections for the change in ecosystem carbon in both scenarios
 - Table 24 and 25 (Appendix 3) summarizes the ecosystem carbon stocks by pool over time in both scenarios, respectively
 - Appendix 5 for supporting data files
 - The accompanying MS Excel file “Darkwoods Carbon Model v.8.7a”, which shows the calculation of the total change in emissions and ex-ante projection of Verified Carbon Units (VCUs)
- Darkwoods VCS Monitoring Report Period 2008-2010 (3GreenTree, 2011), Version 1.4. Verified April 21, 2011 (Rainforest Alliance, 2011b). This initial monitoring period report covers ex-post calculations of starting carbon stocks, the changes to ecosystem carbon stocks in both scenarios, and the calculation of VCUs for the project years 2008, 2009, and 2010.
- Darkwoods VCS Monitoring Report Period 2011-2012 (3GreenTree, 2013). Projected to be Verified in 2013, this Monitoring Report will include ex-post calculations for the project years 2011 and 2012. This Monitoring Report includes significant Darkwoods forest inventory, carbon plot network, and modeling updates that refine carbon inventory calculations for the project in the second verification period.

G1b. Community Information

G1.5 Project Area Communities:

The Darkwoods project area is the private, fee simple property boundary as shown in Figure 4. There are no communities, indigenous people or other public communities or ongoing public use for income,

livelihood, or critical cultural values by any people living or dependent on resources within the project area that would be affected by project activities³.

As typical on many private timberland properties in BC, access is strictly controlled using gates on all road access points. There is no evidence of ongoing illegal access to the property, and NCC staff monitor the property on an ongoing basis from their near by field office near Nelson, BC.

There are 33 non-permanent recreational cabins/trailers with associated lake moorages located on a small area on the eastern edge of the property along the lake, known as Tye. NCC has permitted a continuation of the pre-existing and long-standing seasonal recreational cabin usage through a process of annual rental agreements with the occupants. NCC retains a year-round caretaker to oversee and care for the Tye recreation area. The annual rental net income from these properties is immaterial (<\$20,000/year), and therefore not considered explicitly within the project documentation. Further, there is no change in management of the Tye area in the baseline versus the project scenario, and therefore there are no material net carbon, community, or biodiversity impacts from the project from the Tye area in comparison to the baseline conditions.

There are a few (2-4 active) small (3-4 workers) exploration and mining/prospecting operators occasionally active on the property (but no productive mines). These operations have a minimal surface impact (<1 hectare) and their impacts on carbon emissions are immaterial. Further, there is no change projected for the activities of these mining operators in the baseline versus the project scenario, and hence there are no material net carbon, community or biodiversity impacts from the project versus the baseline.

In general, a limited set of people from the local communities have interacted directly with the Darkwoods property area on an intermittent basis via:

- Seasonal or project-based contracting opportunities for local road construction, maintenance, and logging contractors,
- Project based forest management consulting for inventory, planning, monitoring, or other project related work
- Backcountry recreational use
- Researchers undertaking scientific studies

Project Zone Communities:

As a developed and wealthy geography within British Columbia, the socio-economic conditions in the general region around the project area are stable and diversified, with very well developed infrastructure and integrated economies throughout the southeastern B.C. region and generally across the province itself. The Darkwoods project area is one of many private and crown land timberland properties within the region, and it is not an unique area of special interest or area related to community or stakeholder dependence or livelihood.

For the purposes of the Darkwoods CCB project, a project zone can be based on the surrounding B.C. Ministry of Forests Kootenay Lake Timber Supply Area (Kootenay Lake TSA) as a reasonable administrative boundary within which to consider the impacts of the project on the surrounding communities (Figure 6). The Kootenay Lake TSA is a 1.2 million hectare area that includes the entire project area, along with the main local communities of Nelson, Creston, and Kaslo. This project zone includes a mixture of municipal area, private land, crown timberland, Provincial Parks and Wildlife

³ See also: Choquette, W.T. (2010)

Management Areas, and encompasses the entire project area. The project area is centrally located within the southern end of the TSA (within the large yellow area on Figure 6). The BC MOF reports the Kootenay Lake TSA having a total population of 34,000 in 2006 (BC Ministry of Forests, 2008b).

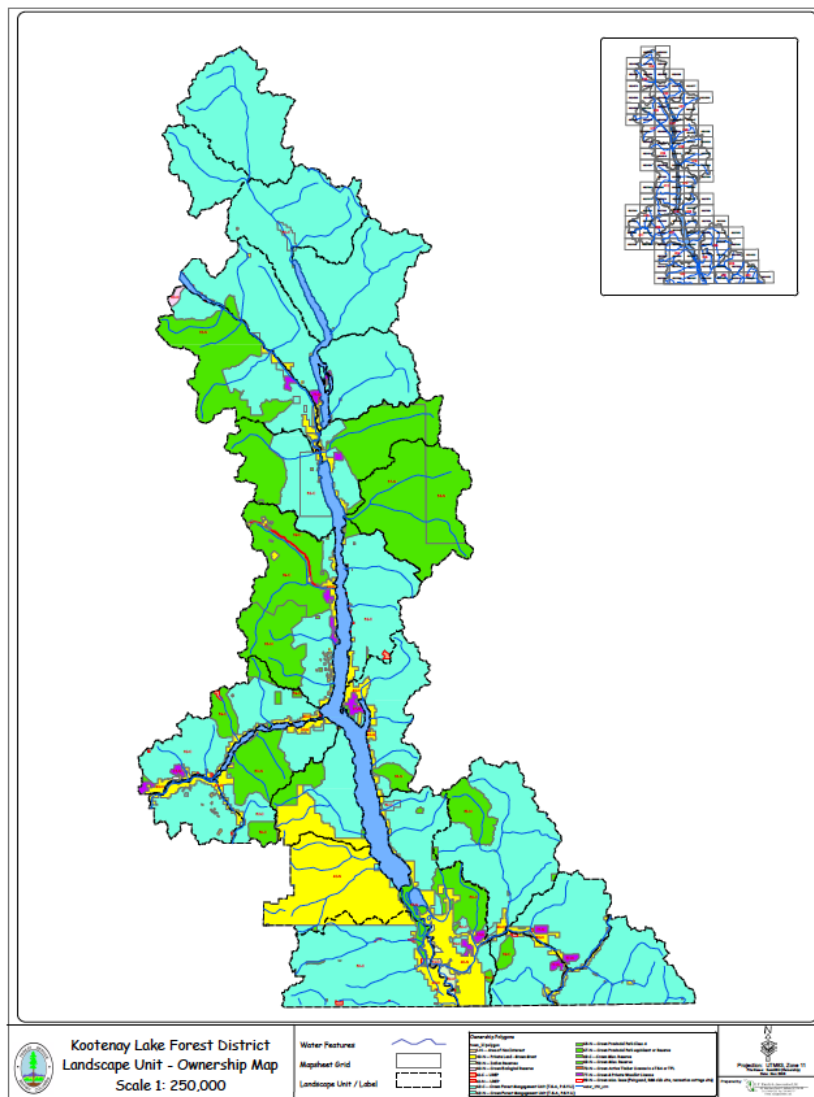


Figure 6 - Kootenay Lake TSA Map with Land Ownership (Project Zone Map)

Project Zone Economic Conditions

The southern interior of B.C. is well developed, with efficient labour markets, a fully developed modern infrastructure and transportation network, excellent private property and forest law enforcement, and robust diversified local economies. The major municipal centers immediately relevant to Darkwoods include Creston (pop. 5,306), Nelson (pop. 10,230), and Kaslo (pop. 1,026).

The South-Central Kootenay area is one of the most economically diverse regions in British Columbia. The Kootenay Lake TSA has a diversity index of rating of 70 (Table 3), making it one of the more diverse economic areas in the province (noting that the slight reduced diversity rating in comparison to the surrounding areas (Figure 7) is due to a high dependence on government services, not forestry).

Table 3 - Southern Interior Forest District Economic Diversity and Forest Vulnerability Indices (BC Stats, 2009a)

Southern Interior Forest Region		
100 Mile House	70	35
Arrow Boundary	74	19
Cascades	72	29
Central Cariboo	66	51
Chilcotin	68	39
Columbia	75	27
Headwaters	64	60
Kamloops	74	11
Kootenay Lake	70	10
Okanagan Shuswap	74	7
Quesnel	55	104
Rocky Mountain	77	12

The Forest Vulnerability Indices in this table are calibrated to those developed for the provincial local areas defined in the main report on this project. As discussed in that report, the most vulnerable area (Quesnel) and least vulnerable area (Victoria) were set to 100 and 0, respectively.

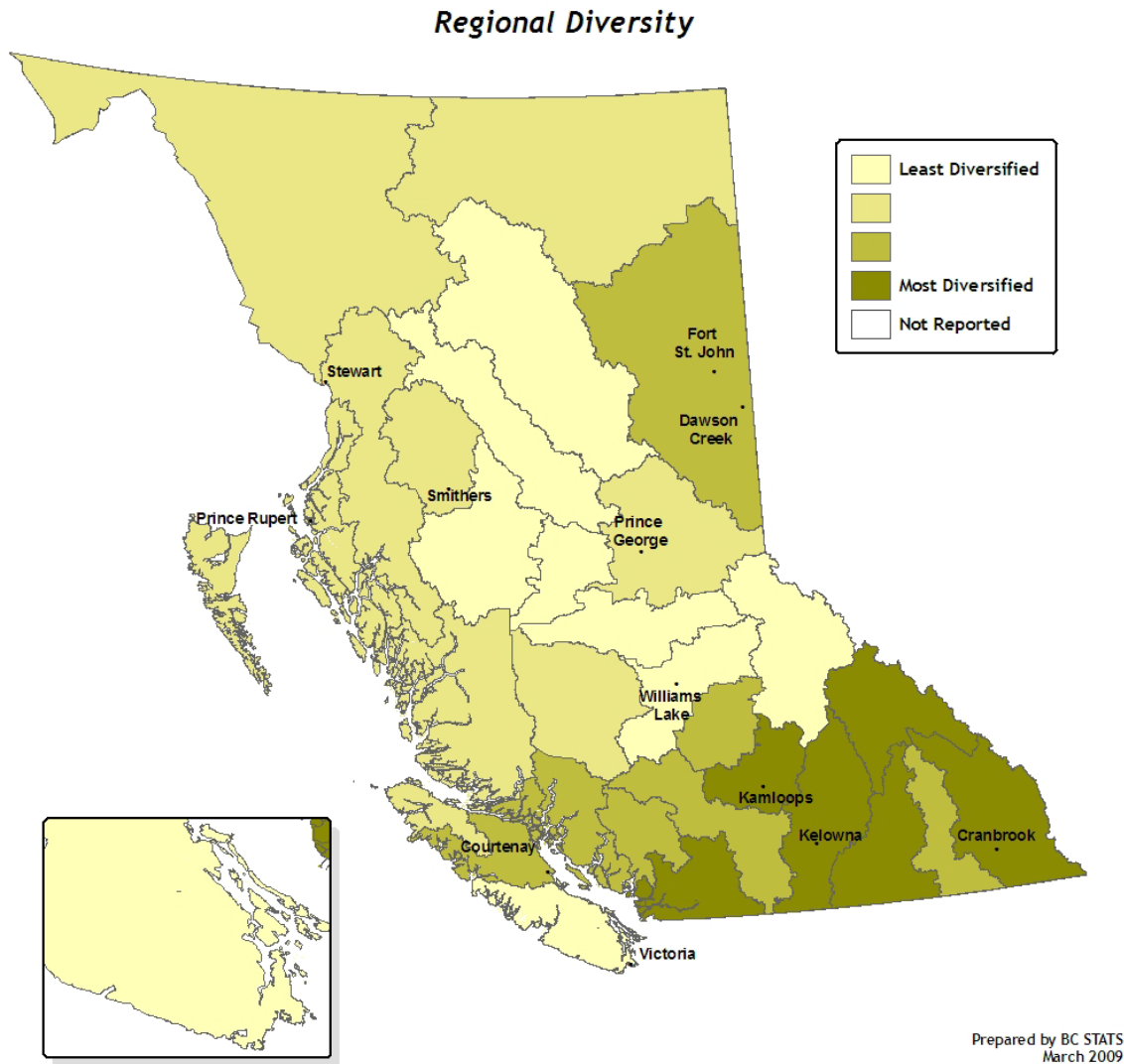


Figure 7 - BC Regional Economic Diversity Rating Map

Historically the area's economy was built on natural resource development (primarily mining and forestry); however, the region has been strongly diversifying economically over the past few decades. The labour force in the project zone has been growing (29% increase between 1986 and 2006), and the forest industry has experienced notable declines (-415 jobs between 1986 and 2006), while construction, other manufacturing, and service-producing industries have been growing (adding 3,746 jobs over the same period) (SIBAC, 2009).

BC Stats (2009b) reports Nelson having a 9% income dependency on forestry and wood processing (as compared to, for example, a 28% public sector services dependency for Nelson), and Creston at 8% (Table 4). The City of Nelson reports 1.2% of the town's labour force (2006 Statistics Canada data) is involved in Forestry, Logging, & Support Activities (Nelson Chamber of Commerce 2012). On a scale of 1-100, Nelson has a Forest Vulnerability Index of 12, and Creston = 11, which indicates these project zone communities are among the least dependent on the forest industry of any forest region in B.C. (BC Stats, 2009b). At the project zone scale, the Kootenay Lake TSA has a Forest Vulnerability Index of 10 (Table 3) (BC Stats, 2009a).

Dependence on Forestry and Wood Processing

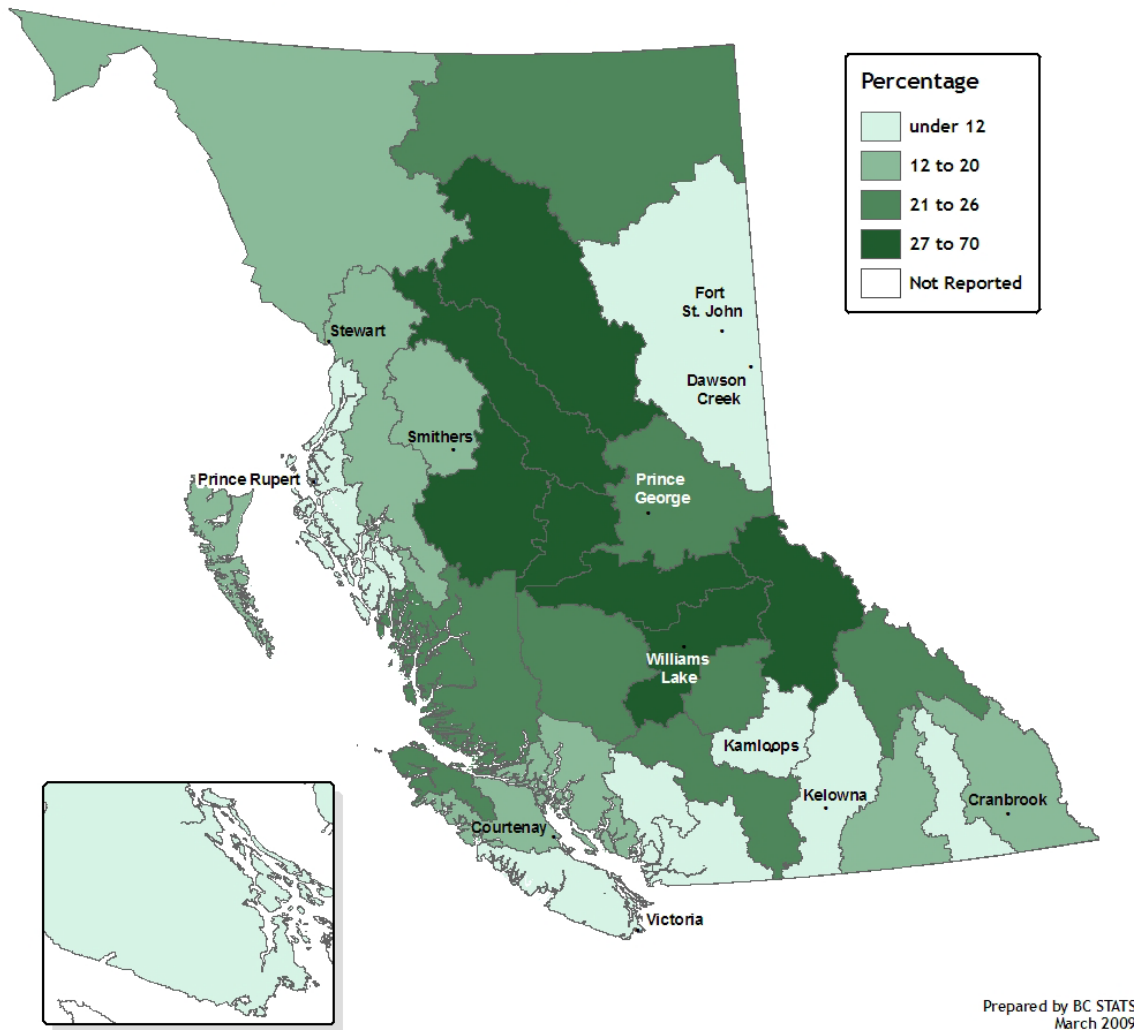


Figure 8 - Forestry and Wood Processing Dependence by Forest Region in BC

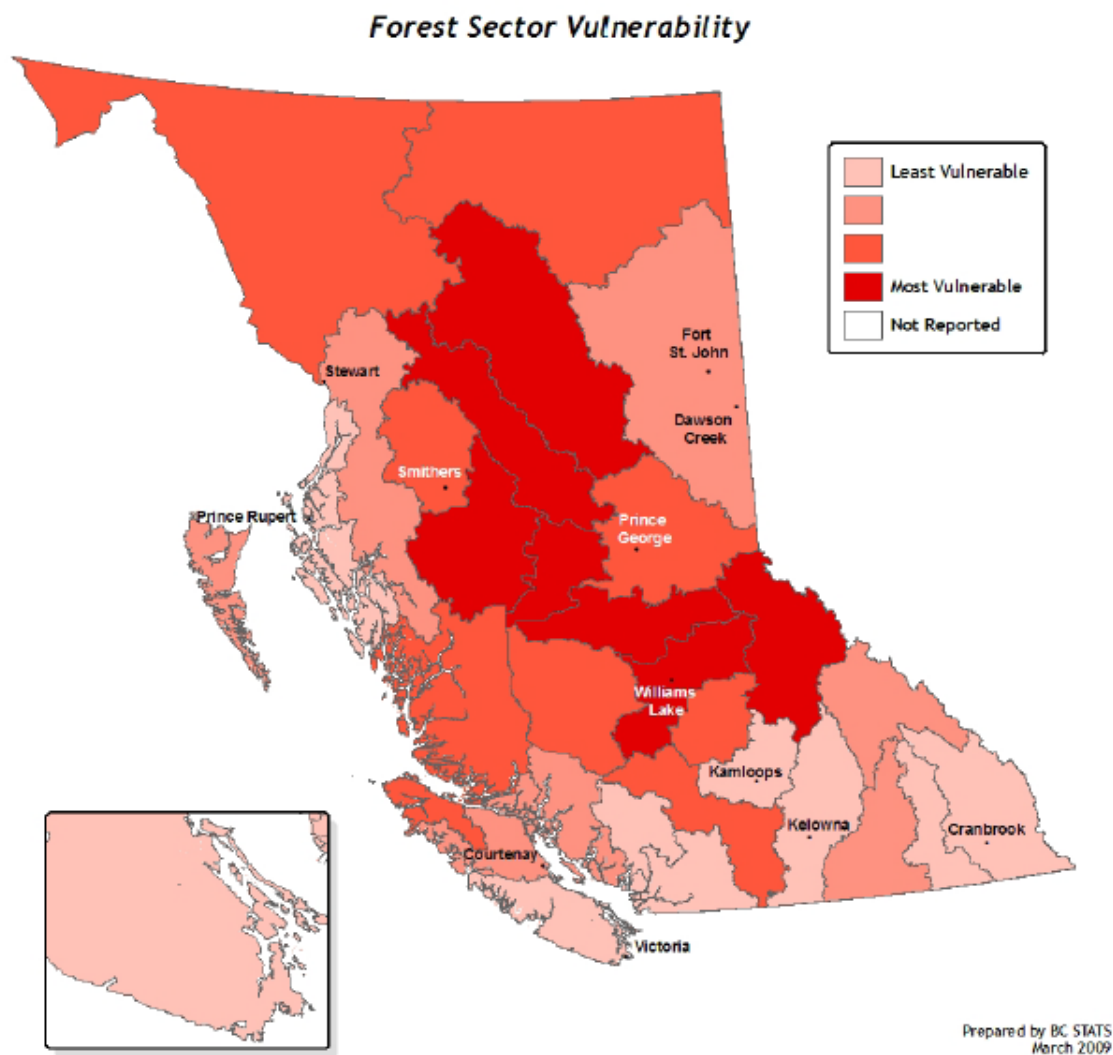


Figure 9 - BC Forest Sector Vulnerability Rating

Figure 9 shows the relative Forest Sector Vulnerability Rating for BC, which indicates the project zone area has a “least vulnerable” forest sector vulnerability rating. This indicates the project zone area is not highly dependent on the forestry sector, but also has a low vulnerability to disruptions in the forest sector.

Table 4 - Select BC Community Income Dependencies by Economic Sector

Percent Income Dependencies (After-Tax Incomes, 2006)

	Forest	Mining & Min Proc	Fish- ing	Agric. & Food	Tourism	High Tech	Public Sector	Const	Film Prod	Other	Trans. Pay- ments	Other non-emp inc
KOOTENAY												
40 Fernie	6	44	0	0	8	0	13	6	0	2	11	9
41 Cranbrook-Kimberley	12	6	0	1	7	0	26	9	0	7	17	16
42 Invermere	13	12	0	2	16	0	16	13	0	3	11	13
43 Castlegar-Arrow Lakes	23	4	0	1	5	1	22	9	0	4	17	14
44 Nelson	9	2	0	1	7	4	28	10	0	5	18	17
45 Creston	8	5	0	7	4	0	19	6	0	1	26	24
46 Grand Forks-Greenwood	23	2	0	4	3	0	17	7	0	4	24	18
47 Trail-Rossland	4	19	0	0	4	1	24	6	0	6	17	18

Scale Context of the Darkwoods Property

For further context, although the 55,000 ha Darkwoods property is the largest private ownership in the project zone, the project area makes up only 4.6% of the 1.2 million hectare Kootenay Lake TSA project zone area. Considered more widely, the Kootenay Lake TSA project zone is a relatively small area within the surrounding productive forest region: the Kootenay Lake TSA makes up approximately 4.8% of the Southeastern BC Forest Region, which covers 25 million hectares, of which 60% is managed forest land, and generates 32.91 million m³ of crown land harvest volume annually. In other words, on an area basis, the Darkwoods project area is a relatively small component of the project zone, which is then a small component of the regional managed forest area.

On a volume basis, the crown land component of the Kootenay Lakes TSA has an annual allowable cut of 640,000 m³/year (effective 2010, <http://www.for.gov.bc.ca/hts/tsa/tsa13/>), an assigned cut level that has declined from 900,000 m³/year in 1990. In 2008, the harvest level from all land ownership sources in the Kootenay Lake TSA was 595,000 m³, however this was a depressed market year – the 2004 total harvest level was 910,000 m³ (BC Ministry of Forests, 2009). Historically, the Pluto Darkwoods property has contributed a marginal volume to the Kootenay Lake TSA harvest volume (i.e. 55,000 m³/year of 595,000-910,000 m³/year = 6-9% of the project zone supply). Considering the surrounding 2 TSA's, where >50% of this harvest volume is delivered, the total harvest level for the Arrow/Boundary TSA, Kootenay Lake TSA, and Rocky Mountain TSA has ranged from 3,761,000 m³ to 5,760,000 m³ annually (B.C. Ministry of Forests, 2009), of which the historical Darkwoods production volume would entail <2% of the local supply. The impacts of the production management changes under the baseline and project scenarios is analyzed further below; however, in general the Darkwoods project area does not play a major role in the overall timber supply market (<8% in all scenarios), although certain aspects of the baseline scenario can increase the proportion of the project zone timber market on a temporary basis.

Forestry Related Employment in the Project Zone (Kootenay Lake TSA):

Lefebvre (2008) undertook a large industry survey of harvesting, silviculture, and primary processing employment in the Central Kootenay Region. Averaged over the three years studied (2005, 2006, and 2008), this report shows the following key forestry sector employment data for the Kootenay Lake TSA (the project zone):

Table 5 - Key Harvesting Employment Variables Within the Project Zone (Lefebvre, 2008; SIBAC, 2009)

Average Harvested Volume (Kootenay Lake TSA; all ownerships)	754,953 m ³ /year
Estimated Direct Harvesting, Planning, and Silviculture Employment (Kootenay TSA)	258 PY (Person Years)
Estimated Direct Primary Processing Employment (Region wide; generated by Kootenay Lake harvest volume)	494 PY (Est. 249 PY within Kootenay Lake TSA)
Total Estimated Direct Forest Industry Employment	846 PY
Employment Coefficients:	
Harvesting/Silviculture (employment generated in Kootenay Lake TSA)	0.342 PY/1000m ³
Primary Processing (employment generated in Kootenay Lake TSA) ⁴	0.327 PY/1000m ³
Total employment generated in Kootenay Lake TSA from Kootenay Lake TSA harvest volume	0.669 PY/1000m ³
Average employment income from forestry and wood processing in the Kootenay Lake TSA (\$2005)	\$42,630/year

Comparatively, a report by SIBAC (2009) reported Statistics Canada data that the Kootenay Lake TSA had a labour force of 585 in Logging and Forestry, and 425 in Wood Products, which are relatively similar to the Person Year based calculations reported in Table 5.

Direct Employment on Darkwoods

There were certain stakeholders with direct employment interests in Darkwoods at the time of project implementation:

At the time of acquisition there were 3 full time employees on Darkwoods. These employees were retained by NCC, and became part of a staffing plan that mitigated impacts to their employment as the project was implemented.

Darkwoods utilized a series of contractors to provide services for various aspects of property management and operations (i.e. timber harvesting, hauling, road construction, road maintenance, forest engineering/layout, inventory, GIS, etc.). None of these contractors had long term contracts or employment agreements that were affected by the project implementation. As typical for the industry, these contractors operate on short-term competitive bid contracts, with no ongoing assurance of activity on the Darkwoods property. These contractors are by necessity mobile within the industry and region, and fluctuations in activity levels within the industry and for any given property owner client are a normal course of business. Hence, although these contractors do have an economic interest in the

⁴ Lefebvre (2008) reported an employment coefficient of 0.655 PY/1000m³ for primary processing in the region, and noted it was difficult to isolate a specific TSA coefficient due to the amount of log trading undertaken, particularly between the Kootenay Lake, Arrow, and Columbia TSA's. In order to look more specifically at employment within the project zone, we have estimated Kootenay specific employment by extrapolating data from the BC Fiber Flow reporting (BC Ministry of Forests, 2009), which used scaling data showed 50.4% of the Kootenay Lake harvest volume is processed within the Kootenay Lake TSA: Primary processing coefficient (Lefebvre, 2008) 0.655 PY/1000m³ * 50.4% (BC MOF, 2009) = 0.327 PY/1000m³

level of activity on the Darkwoods property in terms of business opportunities, none are solely dependent on Darkwoods for economic opportunity.

Local wood processing facilities have acquired logs from the Darkwoods property as market purchases since the property began forest management. However, no long term log supply agreements or other type of permanent contractual relationships were affected by the property acquisition and project implementation. These are market log buyers that acquire supply from a large regional log market basket on a negotiated market sale basis, and such are subject to variations in log supply across a large range of forest operators and owners as a normal course of business. As such, although they do have an interest in the level of log supply from Darkwoods, none are directly dependent on Darkwoods.

The key point is that although certain stakeholders such as contractors and mills have an interest in the level of economic opportunity provided by Darkwoods, there are only the very few stakeholders directly employed at Darkwoods that are dependent on Darkwoods for their current economic livelihood.

Regional Community Values:

The Kootenay's region is renowned for its natural beauty and amenities, which attract a wide range of people and businesses to the area. As an example, Nelson was voted "Best Ski Town in North America" by Powder Magazine in 2012 ([Huffington Post Best Ski Towns](#)). This can also be inferred from the level of economic diversity area (for example, Nelson boasts the highest level of technology industry in the region, which is likely indicative of quality of life decisions rather than any particular technological industry advantage in Nelson). The regional community has also indicated the importance of the environment in recent surveys:

"Based upon two representative public opinion surveys conducted in the Columbia Kootenay region of BC (the first in 2006, repeated in 2012), the following conclusions can be made about environmental values in the area:

- While economic concerns have increased in the Columbia Kootenay region between 2006 and 2012, the importance of protecting and enhancing environmental assets remains strong.
- When asked in 2012 to explain the most valuable assets in the Columbia Kootenay region that contribute to quality of life, three out of four responses included some environmental aspect. The top 4 quality of life assets identified without prompting were "clean air and water", "the beauty of the wilderness", "mountains lakes and rivers in the region", and "outdoor recreation".
- The two most important conservation issues identified by survey participants in 2012 remain the same as in 2006: "water purity/safety" and "protecting against the loss or extinction of local species".
- One belief that is almost universal in the region is an appreciation of the importance of maintaining natural ecosystems, plants and wildlife for quality of life in the area (94% agree)".

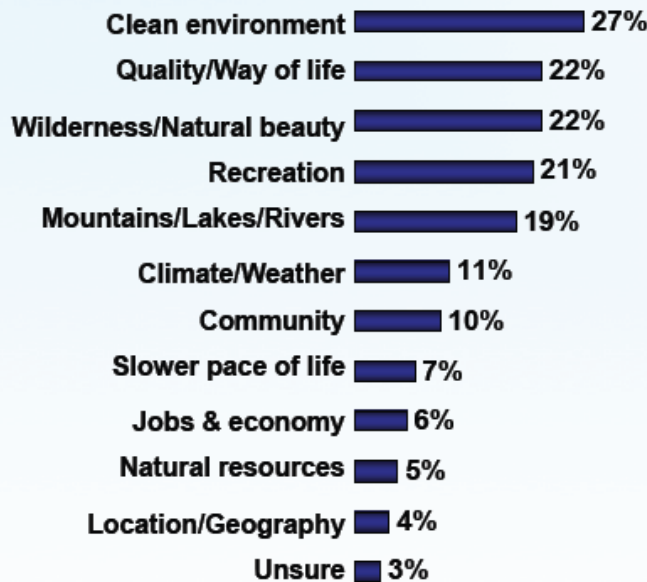
(Source: Cameron, B. 2013)

In addition, Figure 10 shows the rankings related to quality of life from the 2006 public opinion survey (Cameron Strategy, 2006).

Quality of Life Indicators



A2. More specifically, when it comes to quality of life, what do you think is the most valuable asset of the Columbia-Kootenay region that contributes to your quality of life?



Base: All Respondents (n=751)



East Kootenay Conservation Program Study Regional Values and Priorities Study

Figure 10 - East Kootenay Quality of Life indicators

The presence and conservation of large scale properties such as Darkwoods also contributes significantly to local amenity values and livability, which can retain and increase business and employee immigration to the area, and enhance real estate and other values for local community members.

First Nations

The Darkwoods project area lies within the 70,000 km² Traditional Territory of the Ktunaxa Nation, represented most locally by the Yaqan Nikiy or Lower Kootenay Band located near Creston, B.C. The Ktunaxa have ongoing unresolved treaty negotiations with the Government of Canada, and ongoing direct interactions with the B.C. government and most Crown land resource users. Generally, private land property rights have been strongly protected in Canada and respected by First Nations within their Traditional Territories. Private lands are generally excluded from treaty negotiations except when involved on a willing transactional basis. The Ktunaxa are important regional stakeholders who have an interest in cultural and environmental considerations on all areas of their territory, including Darkwoods; however, there is no material level of ongoing use, livelihood activities, or special cultural values on the project area and within the project zone that are materially impacted by the project

activities (i.e. Choquette, 2010). NCC has a strong and active working relationship with the Ktunaxa and is engaged on several regional initiatives of mutual interest.

G1.6 Current Landuse and Related Property Rights

As noted, the Darkwoods project area is fee simple private land with no community property or other surface area interests affecting the project. See Section G1.5 for more detailed description.

G1c. Biodiversity Information

G1.7 Project Zone Biodiversity Description

Current biodiversity within the project zone is very high. For example, to date, 200 animal and 219 plant species have been confirmed on the Darkwoods property (Steeger and Machmer 2009). This includes three IUCN red-list species: Grizzly Bear (*Ursus arctos*)(threatened), Wolverine (*Gulo gulo*)(near threatened), Bull Trout (*Salvelinus confluentus*) (vulnerable). In total, 19 globally, nationally, and/or provincially significant species-at-risk are predicted to occur on the property, 15 of which have been confirmed (Steeger and Machmer 2009). Darkwoods likely provides significant habitat for at least 7 of these species-at-risk, including Grizzly Bear (Special Concern; Blue list), Mountain Caribou (*Rangifer tarandus caribou*) (Threatened; Red list), Olive-sided Flycatcher (*Contopus cooperi*) (Threatened; Blue list), Peregrine Falcon (*Falco peregrinus*) (Special Concern; Red list), Western Toad (*Bufo boreas*) (Special Concern; Yellow list), Wolverine (Special Concern; Blue list) and Bull Trout (Blue list)(NCC 2011).

Biogeoclimatic subzones include the Engelmann-Subalpine Spruce (ESSF) dry-mild (dm) and wet-cold (wc) subzones, as well as the Interior Cedar-Hemlock (ICH) dry-warm (dw) and moist-warm (mw) subzones. Ninety-seven site series (ecosystems within these subzones) have been mapped as part of the baseline inventory on the Darkwoods property. The project zone includes the Creston Valley Wildlife Management Area, which has been designated a RAMSAR site i.e., it contains wetlands are of international significance in terms of ecology, botany, zoology, limnology and hydrology.

Site conditions and natural disturbance from fire, forest pathogens, wind, and other disturbances created a diversity of forest types, many of which are very old (>100 years of age). Man-made disturbances from timber harvesting, fire, roads and historical mining, however, have disturbed the project area over the past 100 years. Threats to biodiversity (see Table 6) were identified by NCC using a method adapted from Harcombe (2007) and assessed using a simple threat rating system developed by The Nature Conservancy, World Wildlife Fund, and Foundations of Success (World Wildlife Fund 2007). The system assigns threat ratings based on an assessment of scope, severity, and irreversibility for each biodiversity target, following IUCN (2006) standards. Since the Libby Dam was built in 1975, hydrologic alterations have eliminated seasonal flooding (important for maintaining wetlands), lowered groundwater levels, and degraded wetland habitats in the region.

Table 6 - Definitions of Threats Associated with Biodiversity on Darkwoods

Threat	Definition
3.2.1 Undersurface Rights Development	Habitat destruction and degradation from Mineral exploration and development
11.1.1 Global warming	Global warming may result in altered hydrological regimes, such as less snow, earlier run-off, etc.
7.1.1 Landscape Disturbance from Fire	Fire that may destroy older forests
5.3.1 Forest Harvesting	Commercial forest-harvesting activities that may result in new roads, animal disturbance, and introduction of noxious weeds
8.1.1 Invasive Species	Habitat loss and degradation from noxious weeds and other invasive alien plants and animals
1.3.1 Recreation Development	The Tye site and other infrastructure used by recreationalists that may displace animals, cause overfishing, or introduce noxious weeds
6.1.1 Recreation activities	Commercial and non-commercial recreation such as snowmobiling, fishing, skiing that may that may displace animals, cause overfishing, or introduce noxious weeds
4.1.1 Roads	Existing and new logging roads may be prone to erosion and slumping, as well as providing access to sensitive areas
8.2.1 Forest health	Forest pests such as mountain pine beetle and other pathogens impact forest health
8.2.2 Caribou Predation	Predators such as wolves and cougars will kill caribou, using expanded road networks for access

G1.8 High Conservation Values

The project zone includes parts of six conservation areas identified in the Canadian Rocky Mountains ecoregional assessment: Salmo River, Slocan River, Salmo/Priest/Selkirks, Kootenay River B, Kootenay River A, and Lower Columbia C (Harcombe 2007). It contains the Canadian portion of habitat for the southernmost herd of Woodland Caribou in North America, a threatened Grizzly Bear population unit, and has the highest tree diversity in British Columbia. It includes the Creston Valley Wildlife Management Area (CVWMA), designated a RAMSAR site (wetlands that are of international significance in terms of ecology, botany, zoology, limnology and hydrology). The last remaining western breeding site for the Northern Leopard Frog (*Lithobates pipiens*) is found in the CVWMA. Large numbers – hundreds of thousands - of waterfowl pass through the valley during spring and fall migration; single day concentrations may exceed 40,000 on occasion. The area supports significant populations of grebes, herons, geese, ducks, and terns. It is the only known breeding site for Forster's Tern in B.C. The marsh supports one of the few breeding population of Black Terns in B.C., which have experienced population declines across their entire range. A continuing loss of habitat due to wetland drainage is the main reason for the decline in Black Tern populations

(<http://www.npwrc.usgs.gov/resource/wildlife/nddanger/species/chlinige.htm>)

Federally listed species in the area include Mountain Caribou⁵(Threatened), Western Screech-owl (*Megascops kennicottii macfarlanei*) (Endangered), Yellow-breasted Chat (Endangered), Badger (Endangered), Western Skink (Special Concern), Giant Helleborine (Special Concern), Short-eared Owl (Special Concern), White Sturgeon (Kootenay River population) (Endangered), Shorthead Sculpin (Threatened), and Columbia Sculpin (Special Concern)(Harcombe 2007). There are at least 20 other plant and animal species on the provincial red and blue list, but which have not yet been assessed by COSEWIC for federal listing. Of special interest is the Creston Northern Pocket Gopher (*Thomomys talpoides segregatus*), a subspecies known only from one valley north of Creston (see Harcombe 2007, Appendix 2 for details).

Landscape Level Ecosystems

Darkwoods is situated in an area of high ridges and mountains interspersed with wide valleys. This area comprises part of the Interior Cedar Hemlock (ICH) biogeoclimatic zone, and encompasses an area of about 5 1/2 million hectares. Commonly referred to as the "interior wet belt" or "inland rainforest", the ICH contains the most productive forests of British Columbia's Interior and more tree species than any other ecological zone in the province. The lower elevation forests are characterized

⁵ The term "Mountain Caribou" used herein refers to the ecotype of Woodland Caribou that occupies southeastern British Columbia. Edmonds (1991) suggested referring to these caribou as the "Mountain/Arboreal ecotype" while Thomas and Gray (2002) referred to them as the "arboreal lichen-winter feeding ecotype. Both are attempts to avoid confusion with the popular name "mountain caribou," which has been applied to caribou occupying mountains in other jurisdictions in Canada

by Western Hemlock (*Tsuga heterophylla*), Western Red Cedar (*Thuja plicata*) and the increasingly rare, Western White Pine (*Pinus monticola*). On drier sites Douglas-fir (*Pseudotsuga menziesii*), Western Larch (*Larix occindentalis*) and Ponderosa Pine (*Pinus ponderosa*) can be found.

Old forests predominate in wetter parts of the zone where fires are infrequent. Favorable conditions mean that trees grow to sizes and ages rivaling the old growth forests of coastal British Columbia. Forests in this zone contain many standing dead trees (snags) and large accumulations of fallen logs and other woody debris. These features of old forests provide valuable habitat for a wide variety of life forms, from seedlings and fungi to birds and bears. Some of the province's best Grizzly and Black bear habitat is located here.

Threatened or Rare Ecosystems

The hydro-riparian ecosystems (rivers, streams, creeks; riparian areas) in the project area are ecologically diverse and face a variety of threats. The size of contiguous segments of these ecosystems has often been restricted by past land use in the region, particularly logging, urban development, dams and hydro development, and pollutants from mining and agriculture. Dry Interior Cedar-Hemlock forests (ICHxw and ICHdw) are a rare forest type degraded by forest harvesting, range activities (cattle grazing), and fire suppression. Past logging has reduced the distribution and abundance of old-growth forests.

Critical Ecosystem Services

The Darkwoods property provides a series of important ecosystem services to the region, including hydrological services for water quantity and quality in watersheds feeding into the Kootenay Lake and the Columbia River Basin, along with other important watersheds to the west. Bull Trout are listed as a species of special concern in B.C. and Darkwoods provides critical breeding habitat for this species in the form of two key drainages (Midge Creek and Cultus Creek). Estimates suggest that more than 12,000 Bull Trout are caught annually in Kootenay Lake. Darkwoods also provides important habitat and refuge for rare and keystone species in the ecosystem, including Caribou, Grizzly Bears, Wolverine, and Western White Pine. The property likely supplies nesting habitat for the native insect pollinators that provide supporting services to agricultural operations in the Creston Valley. Climate-related benefits are accrued from the CO₂ sequestered in the forests of Darkwoods.

Local Community Needs

The Darkwoods property is remote private land with controlled access, and there are no people reliant on the property directly for subsistence or basic needs. The property does not contribute directly towards meeting the basic needs of the local communities, which have the modern, well-developed infrastructure and services typical of the Canadian economy.

Traditional Use

As noted, the Darkwoods project area is a private land is within the Traditional Territory of the Ktunaxa Nation. Although the property may see occasional general use by Ktunaxa members in a manner similar to the general forest region as a whole, there are no known areas within the project area that are critical for traditional use, nor any areas receiving regular, ongoing, or special use (Choquette, 2010). In particular, there are no known areas of traditional use that are affected by implementation of the project scenario.

G2. BASELINE PROJECTIONS

The Darkwoods carbon project considered five potential baseline scenarios following the baseline selection process in VCS Methodology VM0012, which are fully described and analyzed in the Darkwoods VCS PDD, primarily in Sections 2.4, 4.1, and 4.2.

Summarized, the most likely baseline scenario was based on an alternative bidder acquiring the Darkwoods property to implement an intensive industrial logging operation (as compared to the project scenario where NCC acquired the property area for conservation purposes). The activities of the competing acquirer were based closely upon the scenario designed in the independently developed timber valuation in the property appraisal used to value the property for sale (Thrower, 2004). Under this appraisal scenario, the baseline scenario was modeled to deplete existing mature timber from economic stands within the productive and operable land base on the property over a 15 year timeframe; after which harvesting returns to a lower level based on harvesting available maturing timber volume over the remaining project timeline. As modeled, the baseline scenario would harvest approximately 300,000 m³ per year for 15 years, and then revert to a variable harvest level averaging 59,000 m³ per year (ranging from 0 to 165,000 m³ per year) until year 100⁶.

This selected baseline scenario is representative of typical timberland investment financial return expectations at an acquisition price near the asking price, and is consistent with management and operational practices found in adjacent and regional private timberland properties. The baseline scenario's spatial and growth and yield modeling included assumptions consistent with or more conservative than the appraisal analysis and B.C. Ministry of Forests timber supply analysis. And finally, the harvesting operations modeled in the baseline scenario meet or exceed the minimum legal requirements of the B.C. Private Managed Lands Act, local landuse zoning requirements, and is representative of typical operational practice in the region.

G2.1. Baseline Scenario Selection

Multiple baseline scenarios were identified and the most plausible baseline selected as detailed in Section 2.4 of the VCS PDD, and following the requirements of the VCS Standard and VCS Methodology VM0012. Generally summarized, the selected baseline scenario was clearly the most financially attractive scenario, while also being fully consistent with the formal property appraisal timber valuation, typical timberland investment financial expectations, and with regional operating practices on private land.

G2.2. Additionality

Additionality was tested using the VCS Tool for the Demonstration and Assessment of Additionality in VCS AFOLU Project Activities as detailed in Section 2.5 of the VCS PDD. The project was deemed to be additional based upon a Investment Test, wherein a simple financial analysis demonstrated the project scenario conservation activities generate no material net income (actually negative net returns), while the baseline scenario was clearly more financially attractive and capable of generating typical industry returns. In addition, a supplemental Barrier Test and Common Practice Analysis were noted as additional evidence of additionality under the tool.

G2.3. Baseline Carbon Stock Changes

The methods, assumptions, and results of the calculation of carbon stock changes related to the Baseline Scenario are detailed in Section 4.1 of the VCS PDD. Results for the baseline scenario carbon calculations are shown in the VCS PDD as follows: annual harvest volumes in Table 13; growing stock volume in Figure 10; net ecosystem carbon storage for the selected carbon pools in

⁶ However, also note that the results of a major inventory update on Darkwoods in 2011 demonstrated a substantial increase in available timber volume that significantly increases and evens out the ex-ante harvest volume projections beyond year 15 in the baseline scenario, indicating that a higher long term baseline harvest level is available. These data are available, but they do not affect ex-post calculations until after year 15, and will likely not be updated into the Darkwoods VCS PDD ex-ante calculations until a future review date. For simplicity, we will reference the previous ex-ante data from the existing validated VCS PDD only throughout this document.

Figure 11; wood products, manufacturing wastes and logging emissions in Table 14; and net baseline scenario emissions in Table 17.

G2.4. Baseline Scenario Net Impacts on Communities

As there are no people or communities located within the project area, the baseline (without project) scenario would have no direct impact on communities within the project area. However, there are economic, amenity value, and ecosystem impacts from the baseline scenario that would likely have both positively and negatively affected communities in the project zone if the baseline scenario was implemented.

The implementation of the baseline scenario would likely create positive economic benefits from the increased level of timber harvesting, which would provide contracting and services employment opportunities and potentially increased local wood processing employment opportunities. The majority of these economic benefits would occur in the first 15 years of the baseline scenario, when the harvest level is increased, after which a lower and variable long term baseline harvest level occurs for the next 85 years of the project.

Given the highly diversified and low dependence of forestry in the project zone as described in Section G1, the extent and scope of these economic benefits would be limited to those stakeholders engaged in the forest products sector, and the support services thereof. Prior to the baseline scenario, forestry made up a relatively small component of the project zone economies, and a declining source of employment (both over the past 20 years, and projected into the future at declining crown land harvest levels) (Lefebvre, 2008; SIBAC, 2009). The baseline scenario would likely expand the forest sector contribution to the local economy, or recover/delay declines in the sector temporarily over the first 15 years of the project. A substantial portion of this economic opportunity would very likely accrue to stakeholders and communities outside the project zone, as a portion of services are provided by external contractors, a portion of employment by external residents, and a substantial portion of the harvested volume delivered to external facilities.

Simultaneously, the implementation of the baseline scenario would have negative economic impacts on the project zone by foregoing the investments and diversified opportunities in the project scenario, along with significant negative impacts on climate services, other ecosystem services, and regional amenity values that are important to community members in the project zone.

Baseline Scenario Potential Positive Community Impacts:

Once implemented, the baseline scenario would contribute additional forest management and contracting operations on the project area due to the increased baseline harvest levels. An increased level of contractor capacity would have been required to carry out the forest management, road management, harvesting, and hauling operations of the baseline scenario. This economic impact of the baseline scenario can be averaged over the entire 100 year project lifespan, but is also broken out to look at the higher levels of harvest in the first 15 years, then the lower harvest in the following 85 years.

Based on regional industry employment statistics show in Table 5, the increased harvest production in the baseline scenario would potentially result in increased employment, reported in Person Years in Table 7. As an estimate of the economic impact of this increased employment in the forestry sector, Table 7 also estimates the value of the additional employment income added to the project zone.

Table 7 - Select Economic Parameters Related to Forest Sector Activity in the Kootenay Lake TSA (Project Zone)⁷⁸

	Project Average	First 15 years Average	Second 85 years Average
Baseline Harvest Volume	96,000 m ³	307,000 m ³	59,000 m ³
Harvesting Employment (PY)	33	105	20
Processing Employment (PY)	31	100	19
Total Employment	64	205	39
Employment Income (\$/year)	\$2,737,869	\$8,755,477	\$1,682,649

As seen in Table 7, based on project zone employment and income statistics, the implementation of the baseline scenario will provide an estimated 64 jobs on average over the project duration, and an estimated employment income of \$2.7 million per year to the project zone economy. Due to the baseline harvesting schedule, these economic benefits will be front-end loaded during the higher harvest levels in the first 15 years (i.e. 205 jobs over this initial period), then decline for the remaining 85 years (i.e. 39 jobs over the remaining duration).

As noted in Section G1, a portion of the current forestry/harvesting services provided in the Kootenay Lake TSA are provided by contractor employees based outside the project zone (approximately 30-40% additional economic benefit accrues to stakeholders outside the project zone). In addition, some 50.4% of the harvested volume is shipped outside the project zone to be processed in facilities outside the project area (and employing staff from outside the project zone) (Lefebvre, 2008). For the purposes of this analysis, these economic benefits to stakeholders outside the project zone are excluded from Table 7 and this analysis. For reference, at approximately 50% wood export, the additional external wood processing impacts would be approximately equal to the Processing Employment reported in Table 7. External impacts related to harvesting could be estimated to be 30-40% of the Harvesting Employment values in Table 7.

An important caveat to the analysis of economic benefit shown in Table 7 is whether the additional volume harvested in the baseline scenario would be fully additive to the current economy, or whether the harvest volume from the baseline scenario would instead substitute for (i.e. replace) other regional harvest volume, resulting in a substantially lower net economic benefit from the baseline. For example:

- The level of employment in the forest sector has been declining on a fairly regular basis in the project region (see discussion in Section G1), yet timber harvest volumes have fluctuated by as

⁷ See Table 5 for the base data and assumptions used to calculate the values in this table.

⁸ Note that Table 7 reflects only direct economic impacts on workers in the project zone. There is additional indirect employment and induced employment that can be added by use of rough economic multipliers. For the purposes of this analysis, we have chosen to exclude indirect and induced employment for simplicity, due to lack of quality data, and with the assumption that by only considering direct employment/expenditures in both the project and baseline scenarios, we are making an “apples-to-apples” comparison which fairly represents the relative merits of each scenario’s impacts on local communities.

much as 50% annually, which may indicate that forest industry employment in the project zone is not directly tied to annual harvest volume as directly as implied in Table 7. Instead, it may indicate that there is either significant regional mobility to employment in the forest industry, or other significant capacity flexibility that offsets or adapts to harvest volume changes.

- It is very likely that the additional Darkwoods volume harvested in the baseline scenario would substitute for marginal Crown land harvests, lower quality timber,, or higher cost wood, etc. and not create a fully additive economic opportunity for the local forest industry based solely on additional volume as shown in Table 7.
- Local processing facilities may also become more selective in substituting lower cost or higher quality wood instead of simply expanding operations based solely on the new volume from the baseline scenario. This may be indicated by the relative stability of the project zone timber processing industry, despite significant annual regional harvest volume variation (although some of this variation is potentially handled through staffing and shift variability as opposed to capital asset expansion or contraction).
- The extensive log trading and variable log flows within the surrounding region make it difficult to predict what the actual level of harvest export outside the project zone would be. We have extrapolated from the export calculations used in a low harvest year (2008. 595,000m³ all sources) based on available data, however it is not clear what proportion of the local production was exported to other TSA's in much higher Kootenay Lake TSA harvest years (i.e. 900,000 m³ in 2004) that are likely much more reflective of the log supply conditions created by the baseline scenario. It is logical to assume that much higher levels of log export occurred during higher harvest level years, simply because there is no evidence of significant local mill expansion (or contraction) during that time period.

In other words, it is clear that there is very likely not a 1:1 correlation between harvest volume and project zone forest sector activity as shown in Table 7, and evidence suggests there may be significant levels of economic benefit "leakage" that reduce the level of economic benefit created under the baseline scenario. To be conservative, however, we have continued to analyze the baseline impacts as though a 100% of the benefit is received in a direct expansion of the local economy proportional to the baseline harvest volume, when in reality this may be substantially over-estimating the economic benefits to local communities.

Baseline Scenario Potential Negative Community Impacts:

However, in addition to the potential economic benefits, the implementation of the baseline scenario would have had the following negative impacts on project related economic opportunities, recreational opportunities, scientific research, provisioning of ecosystem services and local amenity values of benefit to the local communities:

Climate Services Impacts

As shown within the carbon calculations in this project, the logging in the baseline scenario results in the emission of large volumes of CO₂e into the atmosphere, along with removal of significant carbon sequestration capacity for multiple decades from Darkwoods forests. The near and long term costs of climate change and community adaptation are significant, and when looked at in terms of even minimal carbon pricing as a representation of the cost of GHG pollution to society as a whole, the climate regulating services lost due to the baseline activities are higher than the economic value of the timber operations.

The value of climate services has recently become established through carbon offset markets and carbon tax programs, which can be used to estimate the value of the emissions from the baseline scenario to society. The prices being established on carbon serve to represent the current market for

climate services, and to some degree a proxy for the future cost to society of not acting to avoid climate change. For example, carbon prices can be seen as an alternative to future costs to society for adaptation, and hence provide a legitimate economic impact. In other words, the emissions from the baseline scenario contribute, on a \$/tCO₂e basis to creating future economic costs that local and global economies will have to pay to adapt to. The short term benefits from the logging activity in the baseline should be compared to the economic costs of the climate services impacts of those activities.

As outlined in the Darkwoods VCS PDD, the net change in carbon emissions from the baseline scenario to the project scenario generates substantial levels of emissions reductions, which are tabulated into saleable carbon offsets that serve as a proxy for climate services value to society. Several simple valuation exercises are laid out in Section CM below, which demonstrate the economic impact to climate services based on even minimum carbon pricing models, greatly outweighs the economic benefit of logging in the baseline scenario (as summarized in Table 15).

Other Ecosystem Services Impacts

The extensive logging in the baseline would also very likely have detrimental impacts on other ecosystem services provided to the local communities, even when operated within the legal requirements and recommended best practices, including (but not limited to):

1. Negative impacts on water quantity/timing, water quality/sedimentation, and nutrient loading
 - a. Extensive logging operations fundamentally alter the hydrological regimes within watersheds, as natural, mature forests are replaced by regenerating forests that do not have the same precipitation interception, infiltration, or holding capacity. In particular, water flows can become more variable, and storm events more severe, to the point of damaging stream ecosystems and water quality.
 - b. The expanded area of logging roads, landings, and exposed soil areas after harvesting in the baseline scenario can contribute substantial sedimentation to streams which can negatively affect turbidity and stream habitats.
 - c. The removal of streamside trees can alter stream temperatures, woody debris structure, nutrient loading, and other important stream ecosystem elements. This can occur both in larger fish-bearing streams where multi-tree length buffers are not maintained (i.e. the BC PMFLA requires retention of a few trees per 100 m of large stream length, rather than a solid treed buffer), and in upstream tributaries where buffers are not typically retained.
 - d. Cut over areas tend to release substantial nutrient loads to adjacent streams after harvest through the decay of logging debris, change in surface microclimates, and change in hydrological flows from clearcut areas. This can affect stream nutrients and stream ecosystem behavior.
2. Net negative impacts on wildlife habitat, including both for threatened and endangered species such as caribou and grizzly bears, and potentially for important recreational species such as deer.
 - a. Caribou are highly dependent on old growth, undisturbed forest areas for predator evasion, breeding habitat, and food, among other key habitat features.
 - b. Large predators such as grizzly and wolverine are wide ranging and fairly adaptable to local disturbance, by wide spread habitat and access alteration can have significant detrimental effects.
 - c. Game species such as deer can find improved browse habitat in cutover areas, however the removal of substantial areas of mature tree cover and winter range habitat alteration

- can impact game populations.
- d. And generally, large scale industrial logging in these types of gap dynamic driven forests is a radical change in ecosystem function. Large scale habitat modification will impact the entire natural ecosystem, potentially in ways not fully understood.
3. Likely negative impacts on local native pollinator populations, which reduce pollination in both, project zone crop farming (i.e. near Creston, BC) and non-crop native and non-native plants.
- a. Extensive ecosystem alteration from old forest to young managed forest in the baseline scenario will very likely alter native pollinator habitat and populations.
 - b. The significant economic value of pollination services has been well documented. Research has also shown that highly diverse native bee populations are significant contributors to the pollination and function of both natural ecosystems and agricultural areas. In particular, recent research is showing native bee populations are substituting for declining managed honeybee populations, and are providing an important safety net for agricultural production due to native species diversity.
 - c. Darkwoods is located several kilometers from key agricultural areas, so it is doubtful native bees from the project site contribute directly to pollination services, however it is likely that fully functioning ecosystems on Darkwoods contribute to species diversification, stabilizing closer populations, and providing replacement populations as necessary.

Ecosystem services values are complex and difficult to value reliably, primarily due to the lack of established markets and valuation techniques. It is particularly difficult to measure and attribute ecosystem services provisioning at a specific location and project level. Although several studies have attempted to estimate ecosystem services value using a series of comparative techniques (i.e. Watrecon, 2010; Pagiola et al, 2004; Constanza, et al, 2006; Conservation International 2012; etc.), the ultimate empirical evidence of markets willing to pay for particular ecosystem services values is only starting to emerge (Forest Trends, 2008; Bennett, et al, 2013; Madsen, B. et al, 2011, etc.). Therefore, it is beyond the scope of this report to fully explore estimating ecosystem services values for the Darkwoods property, however there is clearly a significant qualitative value impact from the baseline scenario on ecosystem services provisioning from Darkwoods that is considered in comparison to the traditional economic value outlined above.

Community Amenity Value Impacts

The implementation of the baseline scenario logging would likely have a negative impact on local community amenities and perceptions. Nelson, for example, is a highly diversified community for which recreation and nature are key factors in the lifestyle of living and working in the area. As noted in Section G1, recent regional public surveys have demonstrated the importance of various quality of life indicators to the local communities. The implementation of the baseline scenario may negatively affect the community's perception of its wilderness, outdoor recreation, and aesthetic quality of their surroundings, which may affect the desirability of working and living in Nelson and the project zone. This may then lower the demand (and therefore price) for housing, reduce or reverse the trend of economic diversification, reduce the migration of new businesses locating in Nelson, reduce the growth in tourism, or result in other systemic economic impacts that would affect the overall economy significantly over time. These are complex areas of economic study, and beyond the scope of this report, however there are a substantial number of studies of the impact of local and regional amenity values on real estate, community diversity, and community well-being from the PNW that demonstrate empirical evidence of significant value to communities from being surrounded by wilderness or other natural features. (i.e. Green, G.P., 2005; Garber-Yonts, B.E., 2004, etc.). However, it can be safely concluded there is a high probability that a large scale logging operation on Darkwoods would have detrimental impacts on the quality of life of people in the project zone, including stakeholders in Creston, from which portions of Darkwoods are clearly visible.

Lost Project Scenario Economic Opportunities

The baseline scenario would likely negatively impact the economic opportunities and employment related to the implementation of the carbon project, as described in Section CM. The baseline would also likely negatively affect the economic development and employment related to recreational, scientific research, monitoring, and other activities created by the project scenario. Estimates of these values are projected in Section CM.

Baseline Scenario Community Impact Summary

The implementation of the baseline scenario would have expanded the economic opportunities as compared to the previous owner, a portion of which would have accrued to the communities and stakeholders in the project zone. On average, over the entire project lifespan, the baseline scenario could generate 64 Person Years of new employment (front end loaded with as high as 205 PY over the first 15 years of the project), and contribute \$2.74 million of new employment income into the project zone economy. However, the baseline scenario would also have had negative impacts on the value the communities receive from climate services, ecosystem services, amenity values, and the loss of economic activity related to the project. As detailed in Section CM, these negative impacts exceed the positive impact of the new forest sector opportunity in the baseline scenario.

G2.5. Baseline Scenario Impacts on Biodiversity

The baseline scenario introduces significant threats to biodiversity because the large project area (55,000 ha) constitutes a significant component (12%) of the project zone. The baseline scenario impacts are in the form of (NCC 2011):

1. Undersurface rights development: Habitat destruction and degradation from mineral exploration and development. Mineral exploration could proceed on the project area in a manner largely uninhibited by considerations of its impact on biodiversity.
2. Landscape disturbance by fire: Destruction of rare forest types (old growth, for example) and wildlife mortality. The risk of human-caused fires is increased significantly under the 'without-project' scenario. This impacts both the project area and the project zone.
3. Recreation activities: Unregulated access by recreationalists displaces animals, causes overexploitation (fishing and hunting), introduces noxious weeds (see point 7), and increases risk of wildfire (see point 3). This impacts both the project area and the project zone.
4. Recreation development: Increases area access and usage by humans leading to animal displacement and increased risk of fire. This impacts both the project area and the project zone.
5. Forest harvesting: Commercial forest-harvesting activities may result in new road construction, animal disturbance, and introduction of noxious weeds
6. Roads: Construction, road maintenance, and use of roads results in animal disturbance, introduction of noxious weeds, land erosion and slumping, as well as providing access to ecologically sensitive areas by both recreationalists and predators. This impacts both the project area and the project zone.
7. Invasive species: Habitat loss and degradation from noxious weeds and other invasive alien plants and animals. Invasive species distribution and abundance is often linked to disturbance. Forest harvesting and wildfire, for example, remove significant amounts of forest cover, creating the conditions for secondary succession. This provides an opportunity for invasive species establishment. Roads are the principal corridors for the spread of many invasive species.
8. Caribou predation: Darkwoods provides habitat for the critically endangered South Selkirk mountain caribou herd. Much of the home range is south of Darkwoods and maintaining this

habitat and connectivity is essential to providing refuge from predators. In addition, predators such as wolves and cougars will use an expanded road network (see point 2) for access.

These threats to biodiversity in conjunction with key biodiversity targets are summarized in Table 8.

Table 8 - Biodiversity Threats Related to Darkwoods

↓Threats↓ / ⇒Targets⇒	Dry ICH Forest	Grizzly Bear	Hydro-Riparian Ecosystems	Mountain Caribou	Old-growth Cedar-Hemlock	Rare Ecosystems & Features	Summary Threat Rating
Undersurface rights development	Low		Low	Low	Low	Medium	Low
Landscape disturbance from fire	High		Low	Medium	High	High	Very High
Recreation activities		Medium	Low	Medium			Medium
Recreation development	Low	Medium	Low				Low
Forest harvesting	Low		Low		Low	Medium	Low
Roads		Medium	High	Medium			Medium
Invasive species	Medium		Medium		Low	Medium	Medium
Forest health	Medium		Low	Medium	Medium	Medium	Medium
Global warming	Low		High	High	Low		High
Caribou predation				Medium			Medium
	Medium	Medium	High	Medium	Medium	Medium	High

Some threats are associated with “contributing factors”, which are defined as human-induced actions or events that underlie or lead to one or more direct threats (Table 9). The viability of biodiversity targets is a result of the current state of the property, which in turn is in part a result of past land-use policies and practices. Contributing Factors Associated with Threats to Biodiversity on Darkwoods

Table 9 - Contributing Factors Associated with Threats to Biodiversity on Darkwoods

Contributing Factor	Associated Threats	Notes
Trespass and non-compliance	Recreation activities	Poorly controlled recreation is the major source of impacts associated with this threat
Community commitments	Recreation activities forest harvesting recreation development	Legacy activities are having impacts on biodiversity targets
Economic interests	Recreation activities forest harvesting recreation development undersurface rights development	Operating revenue is generated from activities that are affecting biodiversity targets
Climate Change	Hydrology changes invasive species landscape disturbance from fire	A warming climate will directly impact these targets
Legal obligations	Undersurface rights	Mining exploration must be allowed on private land

G3. PROJECT DESIGN AND GOALS

G3.1. Project Objectives

Fundamentally, the Darkwoods Forest Carbon Project is about the avoidance of logging and property development through the acquisition and management of the property by NCC for conservation purposes. Therefore, the major objectives of the project were to protect and enhance the long-term function of the natural ecosystems on the Darkwoods property. This includes:

1. Protecting and enhancing biodiversity through the protection and management of fully functioning natural ecosystems and key species habitat over time, including:
 - a. Restore and maintain Mountain Caribou habitat and movement areas.
 - b. Restore and maintain hydro-riparian ecosystem integrity, riparian forest structure and function, fish, and hydrologic functioning.
 - c. Restore and maintain stand structure within the range of natural variability for Dry ICH forest.
 - d. Restrict human access to core Mountain Caribou and Grizzly Bear habitat.
 - e. Maintain or restore Grizzly Bear habitat and movement areas.
 - f. Restore and maintain old-forest attributes in old-growth and recruiting cedar-hemlock forests.
 - g. Maintain the integrity of rare ecosystems and habitat features.
 - h. Restore and maintain full suite of naturally occurring species and ecosystems.
 - i. Control noxious weeds and other problematic invasive species.
2. Protecting current carbon stocks and future carbon sequestration potential of the ecosystems across the property, including:
 - a. Reducing timber harvesting to low levels related to long term ecosystem and habitat management, natural disturbance risk reduction, salvage, or other conservation based objectives
 - b. Monitoring property carbon stocks over time
3. Protecting fully functional ecosystems that provide important services and amenities to local communities, while providing diverse economic opportunities and minimizing the overall impact of avoiding the logging and development in the baseline scenario; including:
 - a. Reducing economic impacts by ramping down timber harvests after acquisition
 - b. Protecting the property to retain or enhance ecosystem services provisioning, retaining visual quality and other features contributing to regional wildland amenity values, and allowing limited low impact backcountry recreational and other use opportunities as appropriate
 - c. Supporting research, monitoring, and other diverse economic opportunities across the property, as feasible

G3.2. Project Activities

The key project activity with respect to achieving the climate, community, and biodiversity objectives was the acquisition and conservation of the Darkwoods property to avoid large scale timber harvesting and property development (see also NCC 2012).

Project Activity 1 – Conservation of current forests and ecosystems:

1. Climate impacts: avoidance of GHG emissions resulting from the high level of timber harvesting in the baseline scenario over the project timeline
2. Community impacts: protection of key ecosystem services provisioning related to water quality, air quality, biodiversity, etc. Retention of regional wild land related amenities.
3. Biodiversity impacts: avoidance of habitat modification and operational impacts of logging in the baseline scenario. Retention of large-scale areas of fully functional natural ecosystems and habitat.

Project Activity 2 – Conservation-based small scale timber harvesting:

In order to minimize the direct economic impact of the project on specific community members actively engaged in Darkwoods operations at the time of acquisition, NCC has implemented a harvesting ramp down schedule to move gradually from the previous level of harvest (50,000 - 60,000 m³/year) to the long term conservation management level of approximately 10,000 m³/year.

1. Climate impacts: avoidance of GHG emissions related to larger scale timber harvesting in the baseline scenario over the project timeline
2. Community impacts: protection of key ecosystem services provisioning related to water quality, air quality, biodiversity, etc. Retention of regional wild land related amenities. Retain and provide a set of limited direct economic opportunities for forest products and forest management in the local communities
3. Biodiversity impacts: avoidance of habitat modification and operational impacts of industrial logging in the baseline scenario, including road deactivation. Retention of large-scale areas of fully functional natural ecosystems and habitat.

Project Activity 3 – Carbon project monitoring.

NCC has installed an extensive network of permanent plots to measure the current and ongoing carbon balance of the Darkwoods forests. This plot network is being expanded over time, and plot re-measurements will occur every 5 years.

1. Climate impacts: the activity has no material impact on climate other than as related to successfully implementing the carbon project itself.
2. Community impacts: monitoring activities provide direct economic opportunities to local and regional people.
3. Biodiversity impacts: the activity has no material impact on biodiversity other than as related to successfully implementing the carbon project itself.

Project Activity 4 – Biodiversity project monitoring.

NCC has designed and implemented a robust conservation plan and monitoring program for Darkwoods. The Darkwoods property participates in a series of scientific research projects related to key habitat, mapping, threatened species, and other elements.

1. Climate impacts: the activity has no material impact on climate other than as related to successfully implementing conservation plans on Darkwoods.

2. Community impacts: monitoring activities provide direct economic opportunities to local and regional people. As part of the results of conservation planning, NCC has implemented a public access permitting system, along with a series of user lease agreements and other access planning processes that provide opportunities for community members to interact with the Darkwoods property.
3. Biodiversity impacts: the activity has no material impact on biodiversity other than as related to successfully implementing the conservation management of Darkwoods itself.

G3.3. Project Location

The project area is the entire Darkwoods property, as shown in Figure 1, and as described in Table 1. The project zone is shown in Figure 6. Project activities will occur on a limited basis throughout the project area, as determined by annual project timber harvesting plans and monitoring activities. The specifics of project activities are reported on an ex-post basis within VCS Monitoring Reports.

G3.4. Project Lifetime and GHG Accounting Period

Project Start Date: April 1, 2008

Project Lifetime: 100 years

GHG Accounting Period: 100 years

G3.5. Project Risks

The key project risks are outlined in detail using the VCS risk assessment tools in the VCS PDD, Appendix 1 (the VCS 2007 tool valid at the time of validation) and Appendix 1.1 (the updated risk tool adopted by VCS shortly after validation, and included in the PDD for supplemental purposes). Both risk assessment tools resulted in a very low or low risk rating in all categories.

Although not specifically designed for risk assessments on biodiversity and community objectives, in the case of the Darkwoods carbon project, these VCS risk tools are well suited for assessing natural and human-induced risks for the CCBA project. Fundamentally, the internal and external risks relating to the project design, implementation and management are the same for carbon, community and biodiversity. The natural risks assessed are also similar with respect to biodiversity, community and carbon – since natural risks that might affect carbon in the conserved forests will similarly affect the biodiversity and community values (i.e. ecosystem services provisioning, etc.) being provided by those conserved forests. To some degree, the natural risks may be somewhat more significant for carbon related accounting (wherein an unplanned reversal can have specific quantitative consequences) and ecosystem services provisioning (where a highly disturbed forest may not provide the same level of service), whereas natural disturbances are to some degree a part of biodiversity and fully functioning ecosystems in the long run.

However, in addition to the materials provided in the VCS Risk Assessment Tools, examples of active operational mitigative actions related to risk reduction on Darkwoods include:

1. Property Access - a system of signs and gates are used to define and enforce access zones. Gates are located across roads at property access points, and on mainline roads to restrict access to internal secondary roads.
2. Public Permit System and Access Agreements – NCC maintains a permit system for public access, which is enforced in partnership with the provincial fish and game department. Certain clubs/agencies have special access agreements with NCC, wherein the club self enforces permits, member training, and supervision (for example, NCC has an agreement with the Association of BC Snowmobile Clubs related to Darkwoods access). NCC also maintains

existing lease arrangements with certain commercial backcountry operators, which are valid until 2014, when they will be reviewed to ensure compatibility with NCC property zoning and biodiversity objectives.

G3.6. Maintenance of High Conservation Values

Fundamentally, the high conservation values of Darkwoods are retained by implementing conservation of the forest area on the property. Retaining fully functioning natural ecosystems is the key management activity.

In addition, NCC will implement further measures to reduce the risks to high conservation values:

1. Access - A system of signs and gates are used to define and enforce access zones thereby restricting human impacts on sensitive areas. Gates are located across roads at property access points, and on mainline roads to restrict access to internal secondary roads.
2. Road Deactivation and Rehabilitation - NCC has adopted a position of no net-increase in roads on the property. This means that an equal length of new road deactivation must balance any new roads that are constructed to access forest harvesting opportunities. The objectives of road deactivation and rehabilitation are to support NCC's access management decisions as well as to reduce road-maintenance costs and to reduce threats to biodiversity targets, especially to Mountain Caribou, Grizzly Bear, and hydro-riparian ecosystems. Road rehabilitation serves to remove the road prism and restore native vegetation. Field (2009) developed a road deactivation plan for Darkwoods based on several management criteria, including an estimate of current and future infrastructure requirements and biodiversity threats. The plan was integrated into a strategy for prioritizing management actions related to mountain caribou habitat management (Wilson 2010). Road rehabilitation is significantly more expensive than deactivation and specific projects will be undertaken when resources are available.
3. Ecosystem Restoration/Management – where feasible, NCC may identify opportunities to directly manage or improve key habitats for key species. Although preliminary planning and research has begun, at this time no active restoration projects have been undertaken.
4. Problematic Invasive Species Control - the intensity and extent of noxious weed infestations on the property were investigated and mapped in 2009 (Cronin 2009). Addressing problematic invasive species on Darkwoods will follow priorities set out in Cronin (2009).

G3.7. Post-Project Climate, Community, and Biodiversity Benefits

NCC is one of the largest and oldest conservation organizations in Canada, with a 50-year history of ownership and conservation management activities. NCC's mission and objective is to conserve key properties in perpetuity, which supports the concept that the biodiversity and community benefits will be retained in perpetuity. From a carbon sequestration perspective, the Darkwoods Forest Carbon Project has a 100-year timeline, which is considered permanent in terms of atmospheric CO₂ removals. NCC intends to own and manage the property under the project scenario in perpetuity.

In addition, at the acquisition closing, the Darkwoods property became certified under Environment Canada's Ecological Gifts Program, which provided a tax benefit to the previous owner (as the seller), but which imposes federal government requirements to ensure that conservation related objectives are being met in perpetuity. There are significant penalties for failure to comply with the Ecological Gift Program. Participation in this program further ensures the property will be managed in perpetuity in a manner consistent with the project scenario.

G3.8. Community Consultation

NCC undertook extensive community stakeholder meetings and consultations at the time of the acquisition and the project start date. A list of stakeholder engagements can be found in Section 6 and Table 20 in the Darkwoods VCS PDD.

NCC made community consultation an important focus during their development of the Darkwoods Property Management Plan (PMP) in 2009 and 2010. An important near-term focus of NCC was to communicate the Property Management Plan to key external audiences, in order to build support for NCC's vision and goals for the property, for which a communications plan had been prepared (Field 2009b; Long 2010). Key meetings related to the implementation of this communications plan are included in the VCS PDD Table 20.

NCC will continue to make external communications and community relations a priority. This serves to build awareness, confidence and support for NCC's conservation activities on Darkwoods and elsewhere. Relationships will be maintained and strengthened through regular contact with key stakeholders and partners.

NCC has continued to communicate with key stakeholders since the VCS PDD document was published. For example, NCC met with the local snowmobile clubs on November 14, 2012 to discuss property access and recreational use agreements; and on February 21, 2013 NCC hosted a scientific researchers meeting to discuss research interests, progress, and linking for work related to Darkwoods.

G3.9 Communication of the CCBA Comment Period

NCC has placed a prominent posting on the Darkwoods website with a link to the CCBA comment website.

Due to the limited direct impact from the implementation of the project, NCC has not identified particular affected stakeholders to whom documentation is to be directly distributed, however NCC will be willing to provide documentation should any stakeholders request it.

G3.10 Complaint Hearing and Conflict Resolution

NCC has a formal process to receive communications from stakeholders at local or regional offices, which are documented locally as received. Any significant or unresolved issues are forwarded to the regional office, where records are kept and follow-up plans developed as necessary.

The B.C. Regional Office with delegated authority from the NCC National Office is the final authority for responding to any issues related to Darkwoods.

NCC has received comments related to Darkwoods throughout this process since project initiation, which have resulted in stakeholder meetings and other actions.

G3.11 Project Funding

NCC was able to complete the acquisition transaction using a set of innovative financing mechanisms. By internal policy, NCC typically sets up a permanent endowment fund for each property at the time of acquisition to cover the ongoing costs of property ownership and management in perpetuity. In the case of Darkwoods, net income from the sale of carbon offsets is required to fully finance the Darkwoods Stewardship Endowment Fund.

To date (2013), NCC has sold over 700,000 tCO₂e of VCU's in the voluntary market, the proceeds of which have been used to pay current costs and to build a portion of the target perpetual endowment fund for Darkwoods. NCC has set up a temporary fund of \$1.0 million from carbon sales proceeds to fund all near term carbon offset project costs. The interest from this "sub-fund" is capable of covering

near term project costs, including certification and auditing, monitoring, and other planning and management costs.

In summary, the current income from the sale of carbon offsets has generated income capable of sustaining the minimum ongoing direct carbon project costs in perpetuity. To date, the additional proceeds generated from carbon sales has provided partial funding of the targeted \$12.2 million endowment fund necessary to generate annual income sufficient to cover all property related costs in perpetuity. Proceeds from the ongoing sales of VCUs will continue to generally be directed to the Darkwoods endowment fund and fund biodiversity and conservation efforts by NCC on Darkwoods and beyond.

G4. MANAGEMENT CAPACITY AND BEST PRACTICES

G4.1. Roles and Responsibilities

NCC is the project proponent, and retains sole ownership of the Darkwoods property. NCC employs national, regional, and local staff tasked with oversight or management of various activities related to Darkwoods, along with contract service providers for tasks such as carbon project management, GIS services, monitoring activities, etc.

Organization	Role	Responsibilities	Contact/Address
Nature Conservancy of Canada	Project Proponent	Project ownership and management	BC Regional Head Office #200-825 Broughton Street Victoria, BC V8W 1E5 +1 250-479-3191 1-888-404-8428 (in BC) bcoffice@natureconservancy.ca www.natureconservancy.ca
3GreenTree Ecosystem Services Ltd.	Implementing Partner	1. Project development 2. Ongoing monitoring and implementation support	Mike Vitt 3GreenTree Ecosystem Services Ltd. 1801-71 Jamieson Court New Westminster, BC, Canada V3L 5R4 +1 778-998-5478 mike.vitt@3greentree.com www.3greentree.com

G4.2. Management Team Expertise

NCC has a deep internal and external contract expertise related to biodiversity management, carbon measurement and monitoring, and community engagement. NCC is the largest land trust in Canada, with over 2.6 million acres conserved across Canada. NCC maintains the Darkwoods office in Nelson, BC, where a B.C. Professional Forest Technician manages the day-to-day operations, in conjunction with regional staff based in the Kootenay region. NCC's B.C. headquarters are based in Victoria, BC, while National offices are located in Toronto, ON.

Up to date information related to NCC projects, expertise, and senior staff are available at the NCC website: <http://www.natureconservancy.ca/en/who-we-are/our-team/senior-staff/>

NCC engages highly qualified contract service providers with further expertise in specific areas of carbon project development, ecosystem modeling, Geographic Information Systems and inventory management, field services, etc.

Key Project Staff:

Contact	Company	Title	Project Responsibilities
Tom Swann	Nature Conservancy of Canada	BC Assoc. Regional VP & Director of Land Securement & Major Policy Initiatives	Overall responsibility for Darkwoods, NCC carbon lead team.
Rob Wilson	Nature Conservancy of Canada	Director, Carbon Finance	Carbon project oversight, carbon sales and marketing, NCC carbon lead team.
Nancy Newhouse, RPBio	Nature Conservancy of Canada	Program Manager, Canadian Rocky Mountains Program, BC Region	Regional manager with direct oversight of Darkwoods operations, regional stakeholder engagement, etc.
Dave Wickstrom, RTF	Nature Conservancy of Canada	South Selkirk Project Manager	Darkwoods operational management.
Jane Gilbert	Nature Conservancy of Canada	Chief Communications Officer	NCC national communications.
Mike Vitt, BScFor, MBA	3GreenTree Ecosystem Services Ltd.	Managing Director	Carbon project development, monitoring management.
Clive Welham, PhD, RPBio	3GreenTree Ecosystem Services Ltd.	Managing Director	Carbon project development, monitoring management.
Brad Seely, PhD	3GreenTree Ecosystem Services Ltd.	Managing Director	Carbon project development, carbon modeling, monitoring management.

Additional technical experts provide supporting services for inventory, GIS, and data management; field monitoring activities; biodiversity research and monitoring; and other requirements as needed.

G4.3. Orientation and Training

NCC outlines orientation and training processes in: The Nature Conservancy of Canada Management Policy: Human Resources Policy (2012) and the NCC Darkwoods Occupational Health & Safety Program (2012).

Other specific technical tasks such as carbon monitoring plot installation and measurement are trained at the start of each field season using standard operating procedure documentation.

G4.4. Equal Employment Opportunities

NCC's employments standards are documented in: The Nature Conservancy of Canada Management Policy: Human Resources Policy (2012). NCC is compliant with all federal and provincial legislation in regards to both employment equity and human rights more broadly.

NCC employs locally when feasible. For example, a local person staffs the Darkwoods manager position.

G4.5. Workers Rights

NCC is a national organization fully compliant with the laws and regulations of BC and Canada. NCC employment policies and worker guidelines are documented in: The Nature Conservancy of Canada Management Policy: Human Resources Policy (2012) and the NCC Darkwoods Occupational Health & Safety Program (2012).

G4.6. Worker Safety

NCC's policies related to health and safety, including safe operating procedures, are fully documented in: NCC Darkwoods Occupational Health & Safety Program (2012). Employees and contractors maintain programs that meet or exceed the B.C. Forest Safety Council requirements and WorkSafe BC Regulations.

G4.7. Financial Health

The Nature Conservancy of Canada is Canada's largest not-for-profit land trust organization with over a 50-year history of financial stability. NCC's financial statements are available at: www.natureconservancy.ca/en/who-we-are/annual-reports.

G5. LEGAL STATUS AND PROPERTY RIGHTS

NCC owns the Darkwoods property on a fee-simple basis, and all title and rights documentation is available for review. There are no title disputes on Darkwoods. There are no legal requirements or approvals necessary related to implementation of the carbon project on private land in British Columbia.

G5.1. Relevant Laws and Regulations

This forest carbon project is designed to be compliant with Canadian and British Columbian laws in both the baseline and forest carbon project scenarios.

The Darkwoods carbon project is focused on conserving forest ecosystems and habitat on private lands, and is inherently compliant with provincial, federal, and international laws and regulations that are targeted at regulating development and non-conservation land use.

The primary set of legislation and regulation relevant to forestry activities on Darkwoods is the Private Managed Forest Land Act (PMFLA), which provides reduced property taxes and local land zoning protection in exchange for compliance with minimum forestry operating standards.

As noted in the VCS PDD, the baseline scenario suggests the implementing entity would opt out of the PMFLA registration and revert to lessor local land zoning minimum requirements. However, the modeled baseline scenario has been conservatively modeled to be fully compliant with (or exceeds) all requirements of the PMFLA.

The Darkwoods property is currently registered by NCC in the PMFLA, and hence the project scenario activities are also fully compliant with the PMFLA. Generally, the remainder of the legislation outlined below have no or de minimus operational impact on the baseline or project scenario projections.

An overview of the most relevant Provincial and Federal laws and regulations, which might apply in certain circumstances to the Darkwoods property baseline or project scenario (FSC Canada, 2005):

Provincial Legislation and Regulation:

Statute: *Private Managed Forest Land Act*, S.B.C. 2003, c. 80.

Comments: Sets out the legal framework for forest management on untenured private managed forest land.

Regulations: *Private Managed Forest Land Council Matters Regulation, B.C. Reg. 372/2004. Private Managed Forest Land Council Regulation, B.C. Reg. 336/2004. Private Managed Forest Land Regulation, B.C. Reg. 371/2004.*

Applicability: This law applies as Darkwoods is a privately managed forest and is registered the PMFLA under NCC ownership.

Statute: *Foresters Act, S.B.C. 2003, c. 19.*

Comments: Regulates the forestry profession including registration, certification and discipline.

Provides for the creation and mandate of the Association of British Columbia Forest Professionals.

Applicability: This law applies as Darkwoods has a forest operation and the Forests Act regulates the forest profession. NCC's Darkwoods operations manager is a B.C. Registered Professional Forester.

Statute: *Water Act, R.S.B.C. 1996, c. 483.*

Comments: Among other things, applies to persons making changes in or about streams. See provisions in the Water Regulation that apply to forestry operations, particularly 38(2) 41 and 42.

Regulations: *Groundwater Protection Regulation, B.C. Reg. 299/2004. Water Regulation, B.C. Reg. 204/88.*

Applicability: This act applies to forest operations as it is required that activities around a stream do not negatively impact water quality or wildlife habitat. Generally, the BC PMFLA provides suitable compliance with this act and regulations.

Statute: *Wildfire Act, S.B.C. 2004, c. 31.*

Comments: Regulates the use of open fires in and near forest land. Outlines the government's rights and duties in case of a wildfire.

Regulations: *Wildfire Regulation, B.C. Reg. 38/2005.*

Applicability: This act applies to government response to wildfire on private lands.

Federal Legislation:

Statute: *Fisheries Act, R.S.C. 1985, c. F-14.*

Comments: Among other things, provides that no person shall carry on any work or undertaking that results in the harmful alteration, disruption or destruction of fish habitat, and prohibits depositing deleterious substances of any type in water frequented by fish. Provides for authorizations to alter fish habitat.

Regulations: *Fishery (General) Regulations, SOR/93-53.*

Applicability: This act applies as any work completed on Darkwoods must not result in harmful alteration, disruption or destruction of fish habitat. Generally, the BC PMFLA provides suitable compliance with this act and regulations.

Statute: *Migratory Birds Convention Act, 1994, S.C. 1994, c. 22.*

Regulations: *Migratory Birds Regulations, C.R.C., c. 1035.*

Migratory Birds Sanctuary Regulations, C.R.C., c. 1036.

Applicability: Darkwoods has several endangered bird species that utilize the property during migration periods. This act protects migratory birds and their nests. Generally, the BC PMFLA provides suitable compliance with this act and regulations.

Statute: *Navigable Waters Protection Act, R.S.C. 1985, c. N-22.*

Comments: Among other things, applies to dumping of fill and to structures or bridges on navigable waters.

Regulations: Navigable Waters Bridges Regulations, C.R.C., c. 1231.

Navigable Waters Works Regulations, C.R.C., c. 1232.

Applicability: Darkwoods borders Kooteney Lake and this act applies to activities that impact these navigable waters as well as potentially the larger sections of small streams/rivers that run through Darkwoods. Generally, compliance with the BC PMFLA serves as suitable compliance with these acts and regulations for forestry activities.

International Agreements:

An overview of key international agreements and how or when they may be applicable can be found in (FSC Canada, 2005). No specific agreement is expected to be material to the baseline or project scenarios.

G5.2. Local Authority Approval

As fee simple private land, the title documents accord all aspects related to surface rights to NCC.

G5.3. Other Ownership Encroachment

NCC title documents are properly registered and publicly available through the B.C. Land Title Office. Moreover, all title documents and related plans have been provided to third party auditors and validated/verified as part of the VCS validation/verification of the project. The project area does not contain or involve any areas not owned privately by NCC.

G5.4. Relocation of Affected Peoples

The Darkwoods property is private forestland and does not include any permanent habitation or community activities, and as such the project will not require any relocation or exclusion of communities.

G5.5. Illegal Activities

British Columbia has robust and enforced laws and regulations, with strong private property protections overall. This includes a timber stamping/marketing and scaling reporting system that makes timber theft difficult and rare. In addition, active wildlife poaching monitoring, reporting, and enforcement systems are in place. Timber theft and wildlife poaching are rare in B.C., and small scale if found to occur. There is very low risk of illegal activities materially affecting project carbon stocks or biodiversity. Access routes to Darkwoods are gated and monitored on a regular basis by NCC staff based in nearby Nelson, B.C. There has been no evidence of material illegal activities on the Darkwoods property to date.

G5.6. Carbon Title Rights

NCC is the sole owner of the fee simple title to the Darkwoods property, which includes all surface rights, including all rights to the biomass and related carbon located on the property.

CLIMATE SECTION

CL1. Net Positive Climate Impacts

The climate impacts of the project are fully documented with in the VCS PDD, which demonstrates net carbon emissions reductions from the project activities in each year of the project over the entire project term. The VCS PDD utilizes the VCS VM0012 Methodology. No variations or deviations were made with respect to the climate accounting outlined here.

CL1.1 Net Change in Carbon Stocks

Ex-ante projections of net carbon stock fluxes in the baseline scenario and project scenario, methods, calculations, and data used, and the resulting net change in carbon emissions/reductions are presented in the VCS PDD in the following sections:

- Table 4 – Selection of Carbon Pools
- Table 5 – Emission Sources Included
- Section 3.1 (Table 7) and 3.2 (Table 8) outline data and parameters used in calculation of net emission reductions
- Section 4.1 – Baseline Emission Calculations
- Section 4.2 – Project Emission Calculations
- The results of the Baseline and Project Emissions Calculations can be found in:
- Table 13 – Annual harvest area and volume by scenario
- Table 14 – Harvested wood products emissions and equipment emissions by scenario
- Table 16 – Leakage risk discount calculations
- Table 17 – Projected total emissions and reductions by scenario
- Table 18 – Net emissions reductions, leakage discounts, VCS permanence buffer pool contributions, uncertain factor discounts, and net annual VCU (net emissions reductions) for the project (Table 18 from the VCS PDD is shown below in Table 10):

Table 10 - Final Ex-Ante Net Emission Reduction Calculations (from: Darkwoods VCS PDD, Table 18)

Year	Annual Net Emissions Reductions (tCO ₂ e)	Leakage Risk Discount (tCO ₂ e)	Uncertainty Risk Discount (tCO ₂ e)	Non-Perm. Buffer Contribution (tCO ₂ e)	Cumulative Non-Perm. Buffer Account (tCO ₂ e)	Non-Perm. Buffer Release (tCO ₂ e)	Annual Saleable VCU's (tCO ₂ e)
Rate		Varies	Est. 5%	10%		15.0%	
1	317,823	(36,587)	(14,062)	(31,463)	31,463		235,711
2	331,727	(38,123)	(14,680)	(32,840)	64,303		246,084
3	439,461	(39,709)	(19,988)	(43,600)	107,903		336,164
4	244,640	(43,387)	(10,063)	(24,086)	131,989		167,105
5	350,526	(42,384)	(15,407)	(34,683)	141,671	25,001	283,053
6	436,976	(42,312)	(19,733)	(43,329)	184,999		331,602
7	428,733	(42,528)	(19,310)	(42,502)	227,502		324,392
8	439,273	(42,316)	(19,848)	(43,558)	271,060		333,551
9	401,630	(43,653)	(17,899)	(39,782)	310,842		300,296
10	435,754	(47,233)	(19,426)	(43,163)	300,905	53,101	379,032
11	403,159	(42,913)	(18,012)	(39,942)	340,846		302,292
12	452,612	(46,849)	(20,288)	(44,853)	385,699		340,622
13	413,921	(43,106)	(18,541)	(41,016)	426,715		311,259
14	430,528	(42,425)	(19,405)	(42,683)	469,398		326,015
15	440,911	(42,816)	(19,905)	(43,718)	436,148	76,967	411,439
16	290,837	(23,032)	(13,390)	(28,883)	465,031		225,532
17	131,601	0	(6,580)	(13,183)	478,214		111,838
18	126,403	0	(6,320)	(12,654)	490,869		107,429
19	243,925	(13,649)	(11,514)	(24,273)	515,142		194,488
20	134,467	0	(6,723)	(13,459)	449,311	79,290	193,575
25	127,468	(1,066)	(6,320)	(12,647)	435,666	15,376	122,810
30	116,801	(2,736)	(5,703)	(11,511)	419,237	14,797	111,648
35	158,659	(11,713)	(7,347)	(15,303)	421,388	14,873	139,169
40	97,842	(2,384)	(4,773)	(9,629)	399,101	14,086	95,143
45	81,649	(2,758)	(3,945)	(7,987)	373,181	13,171	80,131
50	117,388	(9,127)	(5,413)	(11,291)	365,191	12,889	104,446
55	119,179	(10,841)	(5,417)	(11,395)	358,843	12,665	104,191
60	90,160	(9,184)	(4,049)	(8,561)	341,401	12,049	80,415
65	74,324	(2,308)	(3,601)	(7,280)	321,131	11,334	72,468
70	41,217	(898)	(2,016)	(4,022)	290,053	10,237	44,519
75	10,230	(1,629)	(430)	(893)	250,340	8,836	16,114
80	75,355	(15,964)	(2,970)	(6,766)	241,545	8,525	58,180
85	139,373	(16,859)	(6,126)	(13,151)	261,206	9,219	112,456
90	107,151	(8,738)	(4,921)	(10,283)	265,729	9,379	92,588
95	113,304	(10,717)	(5,129)	(10,799)	271,766	9,592	96,250
100	83,095	(6,523)	(3,829)	(7,972)	264,882	9,349	74,120
Total	14,660,880	(1,240,247)	(671,032)		(264,882)		12,484,719

CL1.2 Non-CO2 GHG Emissions

As outlined in Table 5 of the VCS PDD, no non-CO₂ GHG emissions are accounted for in the Darkwoods project. Non-CO₂ GHG emissions are de minimis (<5%). This omission is conservative and does not increase the emission reductions claimed by the project.

CL1.3 Other GHG Emissions

As outlined in Table 5 of the VCS PDD, CO₂ emissions related to the combustion of fossil fuels by vehicles/equipment are included.

CL1.4 Net Climate Impact of the Project

The Darkwoods carbon project has a net positive climate impact on an annual basis throughout the duration of the project lifetime. The ex-ante calculations of net climate impacts are outlined in CM 1.1 above.

As seen in Table 10, the Darkwoods project generates a net emission reduction in every year of the project; totaling 12,484,719 tCO₂e of projected net emissions reductions over the project lifetime.

The ex-post net emissions reductions calculations (post-monitoring activities) are outlined in Section 3.4 and within Section 4.2 of the VCS PDD, then calculated and reported for VCU issuance in the VCS Monitoring Reports related to the project.

CL1.5 Carbon Offset Double Counting

Darkwoods does not participate in any other GHG program other than through the VCS and now CCB. Further, Canada did not include forest management accounting in their Kyoto section 3.4 selections, and therefore Darkwoods has no risk of double counting in national Kyoto accounting. Canada has since withdrawn from participation in the Kyoto program entirely.

Darkwoods VCS credits have been validated to be compliant with the B.C. Emissions Offset Regulations, 2008, however this regulation relates to voluntary participation of eligible carbon offset projects in sales to the Pacific Carbon Trust in British Columbia. All offsets related to Darkwoods are registered as VCS VCUs on the Markit Registry. Participation in any potential sales will be in the form of voluntary sales of issued VCUs, and not registered or issued credits in any other form.

CL2. Offsite Climate Impacts ('Leakage')

Offsite leakage methods and calculations are outlined in Section 4.3 of the VCS PDD.

CL2.1 Leakage Sources

As shown in Section 4.3 of the VCS PDD, the project considers leakage from Activity-Shifting and Market Leakage.

CL2.2 Leakage Mitigation

The project has not undertaken any specific leakage mitigation steps other than the imbedded mitigation arising from the harvesting results in the project scenario (wherein the timber production from the project serves to reduce market leakage impacts for the project). As per the VCS Methodology VM0012, the project applies a leakage risk discount factor annually to account for the risk of market leakage from the avoidance of timber harvesting in the baseline scenario.

CL2.3 Leakage Accounting

Leakage accounting calculations are described in Section 4.3 of the VCS PDD. The resulting ex-ante projections of leakage risk discounts projected by the project are shown in Table 16. Ex-post calculations of actual leakage risk discounts applied prior to VCU issuance are shown in each VCS Monitoring Report for the project.

CL2.4 Leakage of Non-CO₂ Gases

Leakage of Non-CO₂ Gases are considered de minimis by the project and excluded.

CL3. Climate Impact Monitoring

CL3.1 Carbon Pools to be Monitored

The data and parameters to be monitored are listed in the VCS PDD in Section 3.2 and Table 8.

CL3.2 Monitoring Plan Development

The climate impact monitoring plan has been developed and implemented on the project, as outlined in the VCS PDD, Section 3.3. Monitoring plan updates and results are described in detail in the project's VCS Monitoring Reports. The results of monitoring are publicly available via the project Monitoring Reports and Verification Reports posted on the VCS Project Database Website.

COMMUNITY SECTION

CM1. Net Positive Community Impacts

As there are no people or communities located within the project area, the project scenario (with project) has no direct impact on communities or stakeholders in the project area. However, there are economic, amenity value, and ecosystem services impacts from implementing the project scenario that both positively and negatively affect communities in the project zone.

Project Scenario Potential Negative Community Impacts:

The implementation of the project scenario would likely have negative economic impacts related to the reduced level of timber harvesting in comparison to the implementation of the baseline scenario. Avoiding the baseline scenario would result in the loss of the economic opportunities described for the baseline scenario in Section G2.4. As noted in that section, given the highly diversified and low dependence of forestry in the project zone as described in Section G1, the extent and scope of these economic benefits would be limited to those stakeholders engaged in the forest products sector, and the support services thereof. Prior to the baseline scenario, forestry made up a relatively small component of the project zone economies, and a declining source of employment (both over the past 20 years, and projected into the future at declining crown land harvest levels) (Lefebvre, 2008; SIBAC, 2009). The baseline scenario would likely expand the forest sector contribution to the local economy, or recover/delay declines in the sector temporarily over the first 15 years of the project. A substantial portion of this economic opportunity would very likely accrue to stakeholders and communities outside the project zone, as a portion of services are provided by external contractors, a portion of employment by external residents, and a substantial portion of the harvested volume delivered to external facilities. As noted in Section G2.4, the lost economic opportunity from the baseline scenario averages an estimated 64 person years of direct employment, worth approximately \$2.74 million in direct employment income to the project zone.

Project Scenario Potential Positive Community Impacts:

However, comparatively, the implementation of the project scenario provides net positive economic, amenity, and ecosystem services provisioning impacts to local communities:

Direct Economic Benefits⁹ – NCC has invested substantial capital into the design, implementation, management, and monitoring of the Darkwoods project. NCC took specific steps to mitigate the potential negative impacts of reducing the harvest levels from the previous owner’s historical logging levels over the first five years of the project to mitigate the direct impacts on specific stakeholders actively working on Darkwoods. NCC has undertaken low volume (and hence high cost) logging operations as planned, along with substantial road maintenance and implemented a road deactivation program. As part of project activities, NCC has done extensive conservation management and biodiversity related planning and field work, extensive carbon project design, development and monitoring work (including an large permanent plot program), undertaken a state of the art new forest inventory, and maintained a full local and regional staff overseeing all operations on the Darkwoods properties. Investments in these activities are significant contributions to employment and economic activity under the project scenario.

According to NCC accounting records, NCC invested approximately \$6 million¹⁰ on the Darkwoods staff, consultants, and materials related to the Darkwoods project over the period of April 1, 2007 (leading up to the project acquisition in April 2008) to June 30, 2012, or approximately \$1.2 million/year on average over the five year period. This spending enabled the hiring of consultants and contractors, the acquisition of equipment and supplies, and paid the wages of NCC local and regional staff for work directly related to implementation of the Darkwoods project. Although the majority of this spending related to services provided within the project zone, some service providers were not based in the project zone (a rough estimate is perhaps 50% of this investment accrued directly to people employed in the project zone).

NCC estimates the ongoing ownership and management costs of the Darkwoods property and carbon project (not including property taxes) to be approximately \$400,000-\$500,000 (averaging \$550,000 over the project duration when including initial year spending), which will be an ongoing level of economic contribution over the entire 100 year project.

In addition, NCC has implemented the low level logging program in the project scenario. Harvest volumes are reported in Table 11. Using the same multipliers from Table 5 that were used in the comparative baseline economic benefit analysis shown in Table 7, the project scenario has the employment and economic impact shown in Table 11.

⁹ To be consistent with the employment coefficients utilized in analyzing the baseline scenario in Section G2.4, we have used economic measures for analysis of the project scenario that are only related to direct spending or employment, with the exclusion of indirect or induced economic benefit multipliers.

¹⁰ These expenses relate to consultants, staff time, administration, supplies and materials related to conservation activities, carbon, and non-logging forestry. It excludes property taxes, capital items, acquisition costs (other than consultant support), logging expenses and revenue, and carbon sales revenue.

Table 11 - Estimated Employment and Income Related to Project Scenario Harvesting

Project Year	NCC Harvest Volume (m ³ /year)	Employment Generated (PY) (Coefficient = 0.669)	Employment Income Totals (\$) (Avg. Income \$42,630/year)
2008	50,763	34	
2009	40,443	27	
2010	23,819	16	
2011	11,913	8	
2012	14,152	10	
5 year average:	28,218	19	\$809,970
100 year average	11,943	8	\$341,040

For comparing employment impacts between the project and baseline scenarios, Table 12 also includes an estimate of the employment related to the NCC expenditures, under the simplistic (and conservative) assumption that the wide range of consultants earn on average the same average income as the average forestry worker in the project zone from Table 5 (likely a conservative assumption).

Table 12 - Estimated Conversion of NCC Expenditures to Employment Impact

	NCC Project Expenditures (\$/year)	Employment Generated (PY) (Avg. Income \$42,630/year)
5 year average	\$1,200,000	28
100 year average	\$550,000	13

Summarized, implementation of the project scenario, at minimum, has created an average of 47 PY (19 PY from harvesting plus 28 PY from NCC project expenditures) of employment over the first 5 years of the project and projects an average of 21 PY per year over the duration of the project. The project contributed approximately \$3.5 million per year of employment income and other expenditures to the local economy over the first 5 years, and is expected to average \$1.36 million per year of employment income and other expenditures for the duration of the project lifetime.

Climate Services Benefits -

The ecosystems on the Darkwoods property provide significant climate benefits in the form of carbon storage and ongoing sequestration in the forest biomass. As calculated in Section CL1, the avoidance of the logging development in the baseline scenario results in significant net emissions reductions over the project lifetime, as a result of reducing GHG emissions into the atmosphere and removing existing atmospheric CO₂ by way of storage in forest biomass. Current and developing carbon market and

taxation schemes serve to establish a real market value on these climate services. This carbon value serves to represent the cost to communities for current and future economic and societal changes due to climate change, and to some degree represents the potential cost of climate adaptation.

By assessing the economic value of the GHG emissions related to the baseline and project scenario under various market pricing scenarios, the economic value of the Darkwoods project to communities can be assessed¹¹. We have selected three representative carbon price data sources to represent a range of climate services benefit valuation¹²:

B.C. Carbon Tax Pricing. The B.C. Government has legally established a carbon tax that prices GHG emissions at \$30.00/tCO₂e, which had been growing annually until a recent temporary price freeze (<http://www.fin.gov.bc.ca/tbs/tp/climate/A4.htm>). B.C. has also passed climate neutral government legislation that sets an carbon offset price of \$25.00/tCO₂e for internal offset pricing via the Pacific Carbon Trust (<http://www.env.gov.bc.ca/cas/legislation/index.html>). For the purpose of this simple analysis, \$25.00/tCO₂e on a real basis is representative of this provincial valuation of climate services.

Voluntary Carbon Markets. Forest Trends (Peters-Stanley, M. et al, 2013) reports that the average market price for forest carbon projects is \$9.20/tCO₂e. NCC has successfully sold VCUs from the Darkwoods project at between \$6.00-\$15.00/tCO₂e. A representative mid-range price of \$10.00/tCO₂e can be used as a market value proxy of the voluntary markets current willingness to pay for climate services at the project level.

California Air Resource Board Cap and Trade Program (CA ARB). A third scenario for the valuation of climate services is the recently implemented CA ARB regulatory Cap and Trade Program. As the most recent very large regulatory carbon market, California ARB allowance prices can provide another comparative real market price on the climate services provided by Darkwoods. In 2013, California ARB Allowance prices ranged near \$14.00/tCO₂e (Point Carbon, 2013).

Table 13 shows a simple valuation¹³ of the climate services provided by the Darkwoods project to communities.

¹¹ Climate services affect the entire global atmosphere, however, for the purpose of this analysis we have assumed the benefits are apportioned to the local communities from which the benefits are realized.

¹² For the simple valuation analysis undertaken here, we are assuming a constant carbon price as a real (sans inflation) snapshot into valuing carbon offsets in order to represent the value of climate services to communities. This simplistic price snapshot approach is conservative, as carbon values in virtually all market conditions are projected or legislated to rise at a rate well above inflation, and actual climate services value from Darkwoods would be significantly higher under increasing price valuation models.

¹³ Note that these simple calculations are intended to be representations of the order of magnitude of climate services value to communities that are suitable for the basic comparative analysis being made in this report. These values are not based upon more complex valuation techniques such as discounted cashflows, and should not be viewed as a representation of actual carbon offset or property asset values on Darkwoods or any other project.

Table 13 - Estimated Darkwoods Climate Services Value to Communities

Net Emission Reductions by Scenario:				
	Baseline (tCO ₂ e/year)	Project (tCO ₂ e/year)	Net Change (tCO ₂ e/year)	Net Saleable VCU's (tCO ₂ e/year)
5 year Average	(335,386)	7,245	1,684,177	253,623
15 year Average	(346,645)	51,200	397,845	308,574
100 year Average	(42,483)	104,126	146,609	124,847
Carbon Pricing Scenarios:				
	BC Legislation	Voluntary	CA ARB	
Avg. \$/tCO ₂ e	\$25.00	\$10.00	\$14.00	
Climate Services Value Estimate Scenarios:				
	Carbon Price (\$/tCO ₂ e)	Baseline Net Emissions (\$/year)	Project Net Emission Reductions (\$/year)	Saleable VCU Volume (\$/year)
BC Carbon Tax Scenario	\$ 25.00	(1,062,063)	2,603,157	3,665,220
Voluntary Market Scenario	\$ 10.00	(424,825)	1,041,263	1,466,088
CA ARB Scenario	\$ 14.00	(594,755)	1,457,768	2,052,523
Average Value Impact:		(693,881)	1,700,729	2,394,610

The project scenario results in an estimated \$1.7 million per year of direct net carbon sequestration services to communities for the duration of the project. When combined with the climate services associated with the avoidance of carbon emissions in the baseline scenario, the project provides an estimated net value of \$2.4 million per year of climate services value to communities.

Project Ecosystem Services Provisioning Benefits:

As discussed in detail in Section G2.4, the project provides substantial further benefits for non-climate ecosystem services provisioning. As noted, these other ecosystem services values are complex and difficult to value reliably, primarily due to the lack of established markets and valuation techniques. It is particularly difficult to measure and attribute ecosystem services provisioning at a specific location and project level. Although several studies have attempted to estimate ecosystem services value using a series of comparative techniques (i.e. Watrecon, 2010; Pagiola et al, 2004; Constanza, et al, 2006; Conservation International 2012; etc.), the ultimate empirical evidence of what markets willing to pay for particular ecosystem services values is only beginning to emerge (Forest Trends, 2008; Bennett, et al, 2013; Madsen, B. et al, 2011, etc.). Therefore, it is beyond the scope of this report to fully explore estimating ecosystem services values for the Darkwoods property; however, there is clearly a significant qualitative value impact from the baseline scenario on ecosystem services provisioning from Darkwoods that must also be considered when comparing to the traditional economic value outlined above. Table 14 outlines a qualitative assessment of non-climate ecosystem services provisioning.

Table 14 - Qualitative Assessment of Non-Climate Ecosystem Services Provision Impacts (see Section G2.4 for discussion)

Ecosystem Service	Baseline Impacts	Project Impacts	Magnitude of Value
Water Quality/Quantity	-	++	Significant ¹⁴
Key Species Habitat	--	++	Significant ¹⁵
Pollinator Services	-	+	Low-Moderate
Recreational Hunting/Fishing	+ ¹⁶	++ ¹⁷	Low to Significant
Outdoor Recreation	-	+	Moderate
Wetland Protection ¹⁸	-	+	Low

As seen in Table 14, the project provides significant community value from non-climate ecosystem services provisioning. Water quality is of particular importance to local communities, due to the project area providing significant watershed flows into Kootenay Lake and at least two major Bull Trout breeding streams contributing significantly into the multi-million dollar Kootenay Lake recreational fishery (Andrusak and Andrusak, 2012). The provisioning of key species habitat is also of particular importance to local communities, as demonstrated (for example) by major government programs focused on regional endangered caribou habitat planning, management, and monitoring. Although difficult to value in monetary terms specifically for Darkwoods, the collective provisioning of these multiple ecosystem services may be on par or exceeding the values calculated for climate services (as shown in Table 13). As a comparative example, Forest Trends (2012) estimates voluntary market water, fisheries and biodiversity market values (\$85 million combined) are approximately 47% of the value of the voluntary forest carbon market (\$182 million/year)¹⁹. If a similar ecosystem services-to-forest carbon market value ration were applied to Darkwoods, the non-climate ecosystem services value could be valued at an estimated \$1.13 million/year for the project duration.

¹⁴ In particular, the Darkwoods property contributes significantly to the valuable Kootenay Lake fishery.

¹⁵ In particular, the endangered mountain caribou are a key species with extensive regional planning efforts underway outside the Darkwoods property.

¹⁶ Some key game species (i.e. deer) may benefit from improved browse habitat created by logging in the baseline scenario, although extensive logging may reduce other habitat attributes for such species. It is unclear whether access for hunting in the baseline scenario would have been allowed, as regional property practices are mixed. To be conservative, a positive impact rating is given.

¹⁷ In some cases, the project would benefit key game species such as deer and elk (browse in harvested areas, winter cover, predation cover, etc.). The benefit to recreational fishery on Kootenay Lake may be significant. NCC has allowed public hunting on parts of the property (whereas the previous owner excluded hunting access).

¹⁸ Significant wetlands are generally protected by minimum legal operating practices, and hence would have minor impacts from either scenario; however the drainages into and out of wetlands and small wetlands may be impacted by logging in the baseline scenario, leading to a negative overall impact rating.

¹⁹ Note that when Forest Trends (2012) includes government-run ecosystem services programs, the water, fisheries and biodiversity markets exceed \$45 billion, demonstrating the eventual value of ecosystem services provisioning to society.

Community Amenity Value Impacts

As noted in Section G1, and Figure 10, the project zone and surrounding region place very high value upon environmental values; consequentially, the conservation of large-scale properties in the project zone can materially contribute to the retention or even improvement of the amenity value of the project zone communities. Amenity values can include local factors that generally contribute to a high quality of life, livability, or desirability of living in the local communities. Studies by the USDA ((i.e. Green, G.P., 2005; Garber-Yonts, B.E., 2004, etc.)), have found that the presence of forests and wild areas can contribute significantly to: community character, cohesiveness, cultural and economic diversity, resource dependence, attractiveness to business, quality of life, and civic leadership. Similar studies have concluded the logical conclusion that communities that are highly desirable can experience higher residential and commercial real estate value, better real estate liquidity, and better community economic conditions (due to factors like attracting diversified new businesses or more talented people, improved community stability/resilience, etc.). Within the project zone, communities such as Nelson can be seen to be building many of these community features over the past several years as the community has moved away from resource industry reliance (i.e. forestry).

Although it is difficult to isolate and estimate a monetary value for the contribution of the Darkwoods project to these amenity values, it can be safely concluded that the conservation of a large scale property proximal to key communities such as Nelson and Creston would contribute to improving the amenity values in the region, particularly in comparison to implementation of the baseline scenario.

CM1.1 Net Community Impacts of the Project

The project has a net positive community impact, based on estimates of the net value impacts of the baseline scenario compared to the project scenario. The baseline scenario provides additional direct employment opportunity (particularly in the initial 15 years); however it is not clear whether all of these employment opportunities will be realized due to timber supply substitution, import of service providers from outside the project zone, or export of the additional timber supply to facilities outside the project zone. The project scenario does provide a lower level of direct employment opportunities, however the project also provides additional employment in fields such as recreation and scientific research on the project area that have not been included here.

As would be expected, the project provides a similarly lower employment income benefit as compared to the baseline. However, the project scenario provides a significant climate services value, which is estimated to strongly outweigh the employment income in the baseline scenario. In other words, the combination of employment income, project expenditures, and climate services provides as much as 1.8 times higher combined economic benefits to the communities in the project scenario as compared to the baseline scenario. And then the project further provides clearly higher net value to communities in terms of the qualitatively assessed non-climate ecosystem services provisioning and in community amenity value.

Table 15 - Comparative Summary of Community Benefit Estimations by Scenario

	Baseline Scenario				Project Scenario				Result
	PY	\$M	# Pos.	# Neg.	PY	\$	# Pos.	# Neg.	
Employment	64				21				3x Higher Employment (Baseline)
Employment Income/Other Spending		\$ 2.74				\$ 1.36			
Climate Services Value		\$ (0.69)				\$ 2.39			
Sub-Total Economic Value:		\$ 2.05				\$ 3.75			1.8x Higher Economic Spending (Project)
Ecosystem Services (net count)			1	-6			9	0	
Amenity Services			0	-1			1	0	
Net Services Impact Total:				-6 (-)				10 (+)	2x Higher qualitative assessment (Project)

In total, the project provides net positive benefits to communities in the project zone.

CM1.2 Community High Conservation Values Impacts

There are no specific High Conservation Values of importance to local communities in the project zone that will be negatively affected by the project. The long term natural conservation of the property will protect or enhance ecosystem services and enhance local amenity, recreation and other values of importance to the local communities.

CM2. Offsite Stakeholder Impacts

The offsite impacts for stakeholders outside the project area, but within the project zone are described within Section CM1.

CM2.1 Potential Negative Offset Stakeholder Impacts

See Section CM1.1.

CM2.2 Offset Stakeholder Impact Mitigation Plans

See Section CM1.1.

CM2.1 Net Offset Stakeholder Impacts of the Project

See Section CM1.1.

CM3. Community Impact Monitoring

CM3.1 Community Impact Monitoring Plan

The primary objective of the Darkwoods community monitoring is to ensure that the project continues to provide the key employment, climate services, ecosystem services provisioning, and community amenity values as described in Section CM.

Community Impact Monitoring Plan:

Tracked annually and reported for each CCBS verification period.

Monitoring Activity 1 - Economic/employment Impact:

Objective: Track and report on NCC economic investment and spending related to the Darkwoods project implementation. Demonstrate that NCC continues to provide diversified economic benefit to the community and region through project related activities and monitoring.

Tasks:

1. Generate and summarize NCC accounting records for expenses related to Darkwoods.

2. Analyze comparative employment value estimates between the baseline scenario and the project expenses during the current verification period.

Monitoring Activity 2 – Climate Services Impact:

Objective: Track and report on the achieved net emission reductions generated by the Darkwoods project in the verification period.

Tasks:

3. Report on the most current ex-ante projections for net emissions from the relevant baseline and project years, and updated relevant average carbon value measures (i.e. B.C. carbon tax and/or offset pricing; CA ARB or Quebec allowance pricing, voluntary market sources, or other justifiable sources).
4. Report on the VCUs projected and issued, along with approximate sales proceeds and prices (to be provided on a confidential basis to verifiers only).
5. Calculate an average estimated climate services value for the verification period.

Monitoring Activity 3 – Other Ecosystem Services Provisioning Impact

Objective: Qualitatively (or quantitatively, if relevant data is available) re-assess the provision of other ecosystem services provisioning by the project in the verification period in comparison to an analysis of the potential baseline scenario impacts.

Tasks:

1. Report on any material project area landuse changes that impact key ecosystem services provisioning
2. Update qualitative value assessments for other ecosystem services in the baseline and project scenario years relevant to the verification period.

Monitoring Activity 4 – Community Amenity Value Impacts

Objective: Qualitatively (or quantitatively, if available) re-assess the provision of positive community amenity value from the project in the verification period in comparison to an analysis of the potential baseline scenario impacts.

Tasks:

1. Report on any material project area landuse changes that impact positive community amenity value.
2. Report on any updated community quality of life or other survey or community information that reports on the value of surrounding area environmental values (for example, references to the importance of water & air quality, recreation, scenery/aesthetics, etc.).

Key Outcome: demonstrate that, on average, the project continues to deliver net positive community benefits in comparison to the baseline scenario by updating the key quantitative and qualitative value assessments made in Section CM1, within each verification period.

CM3.2 Community High Conservation Value Monitoring

Not applicable.

CM3.3 Future Community Monitoring Plan Improvements

The community monitoring plans are currently implemented.

BIODIVERSITY SECTION

B1. Net Positive Biodiversity Impacts

B1.1 Project Biodiversity Impacts

Methodologies to guide biodiversity management are described in the Darkwoods Property Management Plan (PMP; NCC 2011), the latter of which follows the Open Standards for the Practice of Conservation, Version 2.0 (Conservation Measures Partnership 2007) and builds upon the Interim Stewardship Statement prepared for Darkwoods in 2009 (Ennis et al. 2009) and on the Ecological Baseline (NCC 2010, 2011). NCC will employ an adaptive management approach whereby operational objectives and management actions are updated annually based on monitoring outcomes. The intention is to revise the PMP every 5 years to accommodate updated monitoring results and shifting NCC priorities. Recent progress in project activities towards biodiversity management can be found in NCC (2012).

Actions to address threats to biodiversity are organized into five strategies (further details in NCC 2011):

1. **Resourcing:** Funding for management activities will be provided by income from an endowment, lease revenue, project-specific funding, federal government support (short-term) and carbon credit sales.
2. **Zoning and Planning:** The property has been mapped into different zones to protect biodiversity targets by restricting the types of activities that can occur in different areas, such as forest harvesting and public access. Zoning also identifies areas for restoration activities, including road deactivation and rehabilitation, caribou habitat restoration and weed treatments.
3. **Access:** Access to the property will be granted under a public permit system (potentially accessed on-line), club/agency agreements and commercial leases. User responsibilities will vary under these different arrangements, as will the strategies used to enforce compliance with conditions. Fostering relationships with users through outreach is the best way to encourage responsible use of the property.
4. **Management and Practices:** Standard operating practices are currently in place, or will be developed, for data management, forestry-related activities, mineral exploration, restoration methods and fire management.
5. **Monitoring:** Six biodiversity targets will be monitored over time to determine whether goals are being achieved (Table 17). The six targets were adapted from the South Selkirks Natural Area Conservation Plan (Harcombe 2007).

The implementation of these strategies will mitigate or resolve the threats to biodiversity as described for the 'without project' scenario completed in section **G2**. Hence, the difference (i.e., the net biodiversity benefit) from the 'with project' scenario is overwhelmingly positive.

Table 16 - NCC Biodiversity Targets for Darkwoods

Biodiversity Target	Aligned NACP target	Habitat/ Species Type	Ecological Justification	Size	Condition	Landscape Context (LC)	Initially assessed viability ¹	Viability Rationale
Mountain Caribou (<i>Rangifer tarandus caribou</i>)	Mountain Caribou	Mammal	Darkwoods provides habitat for the critically endangered South Selkirk mountain caribou herd	Fair	Fair	Good	Fair	Size: much of home range is south of Darkwoods Condition: forest conditions about 50% at capability LC: well linked to rest of range
Hydro-riparian ecosystems	Hydro-riparian and wetland ecosystems	Rivers, Streams, Creeks – Permanent; Riparian Areas	Rare and ecologically diverse ecosystems that face a variety of threats	Fair	Fair	Good	Fair	Size: often restricted by past land use Condition: logging sometimes has encroached; roads present not always stable LC: all upper reaches and some entire streams creeks protected within Darkwoods
Dry ICH (Interior Cedar-hemlock) forest	Dry Interior Cedar Hemlock forests (ICHxw and ICHdw)	Forest - Temperate	Rare forest type that has been degraded range-wide by forest-harvesting, range activities and fire suppression	Poor	Fair	Poor	Poor	Size: small pieces of original forest left Condition: no major invasive species present LC: fragmented by past land use history
Grizzly Bear (<i>Ursus arctos horribilis</i>)	Grizzly Bear	Mammal	Darkwoods provides habitat for the threatened South Selkirk sub population	Fair	Fair	Fair	Fair	Size: subpopulation uses land outside Darkwoods Condition: variety of successional stages present LC: large landscape well protected and connected to other protected lands
Old-growth cedar-hemlock forest	Old-growth Interior Cedar-Hemlock forests (ICHmw)	Forest - Temperate	Forest-harvesting has reduced the distribution and abundance of old-forest attributes	Poor	Fair	Poor	Poor	Size: remaining stands small Condition: invasive species not a problem LC: fragmented through past logging history
Rare ecosystems and features	none	Variable; Inland Rocky Areas	Rare ecosystems and features require protection from ongoing activities on the Darkwoods property	Poor	Poor	Fair	Poor	Size: variable; sites often unknown Condition: variable; often unknown LC: small areas in large protected matrix

Very Good	Optimal Health: The biodiversity target is functioning at an ecologically desirable status, and requires little management.
Good	Minimum Health: The biodiversity target is functioning within its range of acceptable variation; it may require some management.
Fair	Likely Degradation: The biodiversity target lies outside of its range of acceptable variation and requires management. If unchecked, the biodiversity target will be vulnerable to serious degradation.
Poor	Imminent Loss: Allowing the biodiversity target to remain in this condition for an extended period will make restoration or preventing extirpation practically impossible.
Unknown	Research Need: These targets are known to occur, but information on these criteria is currently unknown.
NA	Not Applicable: This criterion is not significant in assessing the health of this target.

B1.2 High Conservation Values Impacts

NCC BC completed a Natural Area Conservation Plan (NACP) for the South Selkirks Natural Area (SSNA) in 2008 (Harcombe 2007). The Darkwoods property is located within the SSNA. Six primary and 17 nested biodiversity targets were identified for the area, within the context of the Canadian Rockies Ecoregional Assessment. Most of the targets identified for the Natural Area as a whole also occur on the Darkwoods property (NCC 2011). Primary targets include: Mountain Caribou, Grizzly Bear, old-growth cedar-hemlock forests, connectivity, hydro-riparian/wetland ecosystems, and dry Interior Cedar-Hemlock (ICH) forests. NCC's conservation objectives and associated management will address biodiversity threats identified in the SSNACP such that none of the High Conservation Values

identified in the section G1 will be negatively affected by the project (indeed many project activities are designed to enhance these values).

B1.3 Invasive Species

The intensity and extent of noxious weed infestations on the property were investigated and mapped in 2009 (Cronin 2009). Extensive forest harvesting and wildfire provide an opportunity for invasive species establishment, while roads are the principal corridors for the spread of many invasive species. Short-term timber supply from harvesting (to 2013) will decline significantly from the current level of 50,000 m³/year over the next five years. In addition, NCC has adopted a policy of no net-increase in roads on the property. This means that an equal length of new road deactivation must balance any new roads that are constructed. Addressing problematic invasive species on Darkwoods will follow priorities set out in Cronin (2009).

B1.4 Non-Native Species

Non-native species will not be used by the project.

B1.5 Genetically Modified Organisms

GMOs will not be used by the project

B2. Offsite Biodiversity Impacts

B2.1 Potential Negative Offsite Biodiversity Impacts

It is exceedingly unlikely that the project will generate negative offsite biodiversity impacts. The principal objectives of the project are to promote biodiversity habitat and values. The size of the project area (55,000 ha) is such that offsite biodiversity impacts are likely to be positive rather than negative, particularly given the extensive connectivity provided by the project area to other protected landscapes.

B2.1 Offsite Negative Biodiversity Impact Mitigation

Not applicable.

B2.3 Net Biodiversity Impacts of the Project

As per step 1, as a large-scale native ecosystem conservation acquisition, the project is unlikely to cause negative offsite biodiversity impacts. As such, the net effect of the project on biodiversity is overwhelmingly positive. The project area is a component of NCC's South Selkirks Natural Area Conservation Plan (SSNACP; see Harcombe 2007). The SSNACP boundary was predicated primarily on the habitat extent of the Grizzly Bear subpopulation, the extent of the potential and actual range of the South Selkirks Mountain Caribou herd, and the logical boundaries created by three large river systems (Columbia, Kootenay, and Pend Oreille). The boundary includes the east side of the Creston Valley, which encompasses the Creston Valley Wildlife Management Area and the habitat of the endemic Creston Northern Pocket Gopher. Implementation of the SSNACP will result in a large connected network of protected areas totaling over 250,000 acres, containing a reasonably intact and connected mountain range with high habitat diversity ranging from valley bottom riparian and marshes to high subalpine and alpine ecosystems.

B3. Biodiversity Impact Monitoring

B3.1 Biodiversity Monitoring Plan

Monitoring activities and processes on the Darkwoods property will be designed to provide sufficient information to reduce threats and meet biodiversity goals (NCC 2011). Several types of monitoring will play a role on the property:

1. Compliance monitoring: ensure statutory requirements are being met.
2. Implementation monitoring: determine whether management actions were implemented as intended (e.g., deactivating an intended length of road).
3. Effectiveness monitoring (EM): evaluate the ability of management actions to meet goals and objectives (e.g., reducing hydrological risk to achieve water quality objectives).
4. Validation Monitoring: assesses the validity of models. In the case of projections of carbon sequestration and storage rates, a plot network has been installed from which field measures are derived that describe tree growth and productivity. These are used to calculate the relevant carbon storage metrics that are then compared to model output. Field calculations are performed annually and updated on a 5-year interval.

At Darkwoods, effectiveness monitoring and adaptive management are driven by well-defined questions with clear links to management decisions and/or reporting objectives. Key EM questions for Darkwoods were developed concurrently with the PMP process (NCC 2011). A series of team meetings and a structured decision support scheme were used to develop a candidate set of questions/objectives for each biodiversity target (see Table 7). For each target, key questions were developed pertaining to the four potential reasons to undertake EM and related science in support of adaptive conservation management:

1. Address critical knowledge gaps;
2. Resolve action uncertainty;
3. Track improvements;
4. Detect early warning indicators.

Through the EM Scoping process, eight high priority EM questions/objectives were determined to be urgent for implementation (see NCC 2011, Table 12).

B3.2 High Conservation Value Monitoring

An Effectiveness Monitoring Plan for all priority questions was completed in 2011. A set of eight indicators have been monitored to date, as listed in Table 18.

B3.3 Future Biodiversity Monitoring Plan Improvements

The EM planning process will involve continued project team collaboration, expert and partner consultation and possibly hosting an expert workshop. The Darkwoods property has an existing documented monitoring protocol and field plot network. The NCC biodiversity program is extensively documented and outlines standard procedures used across all NCC properties (Durand, 2003). NCC deployed this monitoring program beginning prior to the Darkwoods acquisition closing, and has committed to undertaking annual monitoring activities (NCC 2010, 2011).

Table 17 - Summary of Effectiveness Monitoring - Darkwoods (NCC, 2013)

MUSE Category	Key Attribute	Indicator	Indicator Category	Monitoring Scale	Indicator Rating Thresholds (Viable - Three Stage Index)				Initial Rating (Year)	Current Rating (Year)	Estimated Total (1 = 100%)	Desired Future Rating	Level of Confidence	Monitoring Methodology Action Step	Monitoring Frequency	MUSE Description	Actual MUP Action
					Poor (Low High)	Fair (High)	Good (Medium)	Very Good (Low)									
Targets																	
Mountain Caribou	Forest suitable Mountain Caribou habitat	Proportion of suitable Mountain Caribou habitat for 10 disturbance	Condition	proportion	>10%	5% - 10%	<5%	0				Very Good (8)	Fair	review of forest management plan disturbance	Every 3 years	Less than 2% of suitable Mountain Caribou habitat was not disturbed or undergoing re disturbance	2.1.1, 2.1.2, 2.1.3
Hydro-ecological ecosystems	Water Quality	Number of high risk bridges	Condition	proportion	>1	1 - 1	1 - 2	0				Very Good (None)	Fair	field-based assessment	Every 3 years	All high risk bridges were assessed and high risk areas were identified	
Hydro-ecological ecosystems	Water Quality	Abundance of high risk roads	Condition	proportion	>50%	20% - 50%	10% - 20%	0				Very Good (8)	Fair	field-based assessment, review of forest management plan disturbance	Every 3 years	All high risk bridges were assessed and high risk areas were identified	
Hydro-ecological ecosystems	Water Quality	Proportion of watersheds with up to 10% of forest area	Condition	proportion	<10%	5% - 7%	7% - 9%	>9%				Very Good (>7%)	Fair	review of forest management plan	Every 3 years	Greater than 75% of watersheds with up to 10% of forest area	2.1.1, 2.1.2
Dry FCI forest	Age Structure	Proportion of dry FCI forest	Condition	proportion	<10%	10% - 15%	15% - 20%	>20%				Fair (8%)	Fair	review of forest management plan disturbance	Every 3 years	10% of FCI forest was greater than 140 years old	2.1.1, 2.1.2, 2.1.3, 2.1.4
Grassland	Suitable Habitat	Proportion of forest habitat in managed stage 1	Condition	proportion	<10%	10% - 15%	15% - 20%	>20%				Very Good (>20%)	Fair	review of forest management plan disturbance	Every 3 years	Greater than 20% of forest habitat was in managed stage 1	2.1.1
Grassland	Protection from human-related disturbance	Proportion of forest habitat in managed stage 1	Condition	proportion	0 - 10%	10% - 15%	15% - 20%	>20%	Fair (20%)	Good (40%) (20% - 40%) (20%)	-	Good (80% or greater)	Good	ecosystem management plan	Every 3 years	Greater than 80% of the forest habitat was in managed stage 1	2.1.1, 2.1.2, 2.1.3, 2.1.4
Other																	
Old-growth Cedar-Bradford forest (CBAF)	Age Structure	Proportion of CBAF forest that is greater than 140 years old	Condition	proportion	<10%	10% - 15%	15% - 20%	>20%				Good (>15%)	Fair	review of forest management plan disturbance	Every 3 years	Greater than 15% of CBAF forest was greater than 140 years old	2.1.1, 2.1.2, 2.1.3, 2.1.4
Other																	

GOLD LEVEL SECTION

GL3. Exceptional Biodiversity Benefits

The Darkwoods property is a site of global significance for biodiversity conservation. With a total area of 55,200 ha, the size of the property is significant. The current and long-term biodiversity potential is very high. To date, 200 animal and 219 plant species have been confirmed on the property (Steeger and Machmer 2009). Ninety-seven site series (ecosystems) have been mapped for the property (NCC 2010) for the baseline inventory.

IUCN red-list species found on the property to date are: Grizzly Bear (*Ursus arctos*) (Threatened), Wolverine (*Gulo gulo*) (Near Threatened), Bull Trout (*Salvelinus confluentus*) (Vulnerable). Other notable species: In total, 19 Globally, Nationally, and/or Provincially significant species-at-risk are predicted to occur on the property, 15 of which have been confirmed (Steeger and Machmer 2009).

Darkwoods likely provides significant habitat for at least 7 of these species-at-risk, including Grizzly Bear (Special Concern; blue list), Mountain Caribou (*Rangifer tarandus caribou*) (Threatened; Red list), Olive-sided Flycatcher (*Contopus cooperi*) (Threatened; Blue list), Peregrine Falcon (*Falco peregrinus*) (Special Concern; Red list), Western Toad (*Bufo boreas*) (Special Concern; Yellow list), Wolverine (Special Concern; Blue list) and Bull Trout (Blue list).

The biodiversity significance of Darkwoods is partly a consequence of its geographic extent but is also related to its topographical gradient. From an elevational perspective, the property extends from warm, moist valley bottoms, to cold and dry, high elevation Engelmann spruce/Subalpine fir forests. Under the warmer climate predicted for the region, many lower elevation species are likely able to migrate upwards as their original habitat deteriorates while upslope conditions begin to ameliorate.

In addition to the magnitude of the area encompassed by Darkwoods, the property has considerable importance because of its adjacency to, and connectivity with, three provincial protected areas. These are West Arm Provincial Park (WAPP), Midge Creek Wildlife Management Area (MCWMA), and the Creston Valley Wildlife Management Area (a 7,000 ha RAMSAR site; see s. G1.8). WAPP is an area of about 25,300 ha that extends along the shore of Kootenay Lake from Nelson to Harrop, BC and up to the peaks behind these towns. As the natural habitat for several endangered species, the area was protected for its biodiversity value and which also serves to protect Nelson's water source. The MCWMA is a 15,163 hectare protected area that provides critical lower elevation wildlife habitat for a variety of species-at-risk, including grizzly bear and mountain caribou. In totality, Darkwoods constitutes about 50% by area of a contiguous protected area of about 100,000 ha.

Literature Cited

- 3GreenTree Ecosystem Services Ltd. 2011a. Darkwoods Forest Carbon Project Description Document, Verified Carbon Standard v1.8. Vancouver, B.C. Available Online at: <https://vcsprojectdatabase2.apx.com/myModule/Interactive.asp?Tab=Projects&a=2&i=607&lat=49.348783&lon=-116.786823&bp=1>
- 3GreenTree Ecosystem Services Ltd. 2011b. Darkwoods Forest Carbon Project – 2008-2010 Monitoring Report v.1.4. Vancouver, B.C. Available Online at: <https://vcsprojectdatabase2.apx.com/myModule/Interactive.asp?Tab=Projects&a=2&i=607&lat=49.348783&lon=-116.786823&bp=1>
- 3GreenTree Ecosystem Services Ltd. 2013. Darkwoods Forest Carbon Project – 2011-2012 Monitoring Report v.TBD (Draft - Verification Pending). Vancouver, B.C.
- Braumandl, T., M. Curran, G. Davis, C. DeLong, M. Fenger, M. Ketcheson, D. Norris, B. Peschke, H. Quesnel, C. Steeger, R. Stewart, and G. Woods. 202. A field guide for site identification and interpretation for the Nelson Forest Region. Land management Handbook 20, Ministry of Forests, Victoria, B.C.
- B.C. Ministry of Forests. 1994. Kootenay Lake Timber Supply Area Socio-Economic Analysis. Victoria, BC.
- B.C. Ministry of Forests. 2001. Kootenay Lake Timber Supply Area Timber Supply Review - Analysis Report. Victoria, BC.
- B.C. Ministry of Forests. 2008a. Kootenay Lake Timber Supply Area Timber Supply Review - Data Package. Victoria, BC.
- B.C. Ministry of Forests. 2008b. Kootenay Lake Timber Supply Area Timber Supply Review – Information Report. Victoria, BC.
- B.C. Ministry of Forests. 2009. Fiber Flow Analysis: <http://www.for.gov.bc.ca/het/fibreflow/DKL.htm>. Published online by BC Ministry of Forests, Competitiveness Branch, Fiber and Mill Information. Web accessed May 1, 2013.
- B.C. Stats. 2009a. 2006 Economic Dependency Tables for Forest Districts. Victoria, BC.
- B.C. Stats. 2009b. British Columbia Local Area Economic Dependencies: 2006. Victoria, BC.
- Bennett, G., Carroll, N., Hamilton, K. 2013. Charting New Waters: State of Watershed Payments 2012. Washington, D.C. Available Online at: http://www.forest-trends.org/embargoed/water_2012/sowp_2012_exec_sum.pdf
- Cameron, B. 2013. Kootenay Conservation Program Public Opinion Surveying Excerpt – Implications for Assessing Public Support for Darkwoods. Personal Communication commentary note – February 23, 2013.
- Cameron Strategy. 2006. East Kootenay Conservation Program: Regional Values and Priorities Study. Report prepared for the East Kootenay Conservation Program.
- Choquette, W.T. 2010. Archaeological Overview Assessment for the NCC Darkwoods Conservation Area. NCC Internal Report, December 31, 2010. Yahk, BC.
- Conservation International. 2012. Nature Providers – Ecosystem Services and their Benefits to Humankind. Internal Brochure - http://www.conservation.org/Documents/CI_Ecosystemservices_Brochure.pdf. Arlington, VA.

- Conservation Measures Partnership. 2007. Open standards for the practice of conservation, version 2.0. Foundations of Success, Bethesda, MD.
- Constanza, R., Wilson, M., Troy, A., Voinov, A., Liu, S., D'Agostino, J. 2006. The Value of New Jersey's Ecosystem Services and Natural Capital. Report to the New Jersey Department of Environmental Protection. Burlington, VT.
- Cronin, E. 2009. Inventory of weeds for the Darkwoods property of Nature Conservancy of Canada. Prepared for: Nature Conservancy of Canada, Victoria, BC.
- Durand, R. 2003. Baseline inventory protocol: A guideline for inventorying and mapping protected areas. Nature Conservancy of Canada, British Columbia Region, and Habitat Acquisition Trust.
- Edmonds, E.J. 1991. Status of woodland caribou in western North America. Rangifer Spec. Issue No. 7. pp. 91–107.
- Ennis, T., Klein, J., Herenbrink, J., Harcombe, A., Wilson, S., Field, P., and McGuinness, K. 2009. Darkwoods interim stewardship statement. Nature Conservancy of Canada, Victoria, BC.
- Field, P. 2009a. The Darkwoods property: access management plan. Prepared for Nature Conservancy of Canada, Victoria, BC.
- Forest Trends. 2008. Payments for Ecosystem Services: Market Profiles. Washington, D.C. Available Online at:
http://moderncms.ecosystemmarketplace.com/repository/moderncms_documents/PES_Matrix_Profiles_PROFOR.1.pdf
- Forest Trends. 2012. Innovative Markets and Market-Like Instruments For Ecosystem Services – The Matrix 2012. Washington, D.C. Available online at:
http://www.ecosystemmarketplace.com/documents/acrobat/the_matrix.pdf
- FSC Canada. 2005. FSC BC Guidance - A Companion Document to the FSC-BC Regional Standards for British Columbia. Toronto: FSC Canada.
- Garber-Yonts, B. E. 2004. The economics of amenities and migration in the Pacific Northwest: review of selected literature with implications for national forest management. Gen. Tech. Rep. PNW-GTR-617. Portland, OR: U.S. Department of Agriculture, Forest Service, Pacific Northwest Research Station. 48 p.
- Green, G.P., Deller, S.C., Marcouiller, D.W. (Editors). 2005. Amenities and Rural Development: Theory, Methods, and Public Policy. Edward Elgar Publishing, Inc. Northampton, MA.
- Harcombe, A. 2007. South Selkirks natural area conservation plan. Nature Conservancy of Canada, Victoria, BC.
- IUCN_CMP. 2006. Unified Classification of Direct Threats, Version 1.0
<http://www.iucn.org/themes/ssc/sis/classification.htm> <IUCN-CMP 2006b.pdf>.
- Lefebvre, P. 2008. Central Kootenay Region Timber Harvesting and Processing Employment Survey. Final Report, December 15th, 2008. Vancouver, BC. Available online at:
<http://www.for.gov.bc.ca/het/fibre.htm>
- Madsen, B., Carroll, N., Kandy, D., Bennett, G. 2011 Update: State of Biodiversity Markets. Washington, DC: Forest Trends, 2011. Available at:
http://www.ecosystemmarketplace.com/reports/2011_update_sbdm.
- NCC (Nature Conservancy of Canada). 2010. Darkwoods Conservation Area Baseline Inventory Report, Volumes 1 and 2. Victoria, BC.

- NCC (Nature Conservancy of Canada). 2011. Darkwoods Conservation Area Property Management Plan 2011 – 2016, Victoria, BC.
- NCC (Nature Conservancy of Canada). 2012. South Selkirks Natural Area Conservation Plan – Year 4 Annual Progress Report. Victoria, BC.
- NCC (Nature Conservancy of Canada). 2013. Summary of Effectiveness Monitoring (Darkwoods). Internal Document. Victoria, BC.
- Nelson Chamber of Commerce. 2012.
<http://www.discovernelson.com/images/2012%20Economic%20Development.pdf>
- Pagiola, S., von Ritter, K., Bishop, J. 2004. Assessing the Economic Value of Ecosystem Conservation. The World Bank Environment Department, Environment Department Paper No. 101. Washington, D.C.
- Peters-Stanley, M., Hamilton, K., Yin, D. 2012. Leveraging the Landscape: State of the Forest Carbon Markets 2012. Washington, D.C. Available online at: http://www.forest-trends.org/documents/files/doc_3242.pdf
- Point Carbon. 2013. Market Comment, Carbon Market North America Newsletter, Volume 08, Issue 09, May 17, 2013. http://www.pointcarbon.com/polopoly_fs/1.2378333!CMNA20130517.pdf
- Rainforest Alliance. 2011a. Verified Carbon Standard: Validation Report – Darkwoods Forest Carbon Project. Richmond, VT. Available Online at:
<https://vcsprojectdatabase2.apx.com/myModule/Interactive.asp?Tab=Projects&a=2&i=607&lat=49.348783&lon=-116.786823&bp=1>
- Rainforest Alliance. 2011b. Verified Carbon Standard: Verification Report – Darkwoods Forest Carbon Project, 2008-2010 Monitoring Period. Richmond, VT. Available Online at:
<https://vcsprojectdatabase2.apx.com/myModule/Interactive.asp?Tab=Projects&a=2&i=607&lat=49.348783&lon=-116.786823&bp=1>
- SIBAC. 2009. Southern Interior Beetle Action Coalition Kootenay Lake TSA Socio-Economic Trend Assessment. Merritt, BC.
- Steeger, C., and M. Machmer. 2009. Wildlife inventories on the Nature Conservancy of Canada – Darkwoods property. Prepared for: Nature Conservancy of Canada, Victoria, BC.
- Thomas, D. C., and D. R. Gray. 2002. COSEWIC assessment and update status report on the woodland caribou *Rangifer tarandus caribou* in Canada. Committee on the Status of Endangered Wildlife in Canada, Ottawa, ON.
- Thrower, J., & Orr-Ewing, A. 2004. Appraisal of Timber Value on Darkwoods Forestry Private Timberlands. Vancouver, BC: J.S. Thrower & Assoc.
- Voluntary Carbon Standard. 2008a. Guidance for Agriculture, Forestry and Other Land Use Projects. Washington, D.C.: VCS Association.
- Voluntary Carbon Standard. 2008c. Tool for AFOLU Non-Permanence Risk Analysis and Buffer Determination. Washington D.C.: VCS Association.
- Voluntary Carbon Standard. 2010a. Tool for AFOLU Non-Permanence Risk Analysis and Buffer Determination. Washington D.C.: VCS Association.
- Voluntary Carbon Standard. 2010b. Tool for the Demonstration and Assessment of Additionality in VCS Agriculture, Forestry and Other Land Use (AFOLU) Project Activities. Washington D.C.: VCS Association.

Watrecon Consulting. 2010. Economic Activity and Ecosystem Services in the North Saskatchewan River Basin. Report Submitted to the North Saskatchewan Watershed Alliance. Edmonton, AB.

Wilson, S. F. 2010. Setting priorities for management actions within mountain caribou habitat on NCC's Darkwoods property. Prepared for Nature Conservancy of Canada, Victoria, BC.

Wilson, R. 2013. Personal Communication, May 2013. Manager, Federal Partnerships Program, Nature Conservancy of Canada, Toronto, ON.

World Wildlife Fund. 2007. Resources for implementing the WWF project & programme standards. Step 1.4. Define: threat ranking. World Wildlife Fund, Washington, DC.